

Annex I – JRP protocol

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22RPT01 TracInd BVK-H

Traceability for indentation measurements in
Brinell-Vickers-Knoop hardness

Start date: 01 September 2023

Duration: 36 months

Coordinator
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Glossary

ASTM	American Society for Testing Materials
BIPM	Bureau International des Poids et Mesures
BVK	Brinell-Vickers-Knoop
CCM	The CIPM's Consultative Committee for Mass and Related Quantities. The most senior technical committee within the SI that deals with mass and related quantities metrology fields
CEN	European Committee for Standardization
CIPM	Comité International des Poids et Mesures (International Committee for Weights and Measures). The committee within the Metre Convention that has overall responsibility for the SI
CMC	Calibration and Measurement Capability. Officially recognised capability recorded in a public database. Specifically, those of the NMIs as registered in the KCDB of the BIPM
COOMET	Regional Metrology Organisation of the countries of Central and Eastern Europe: Euro-Asian Cooperation of National Metrology Institutions
COOMET TC-M	COOMET Technical Committee for Mass
DI	Designated Institute. A body that officially acts as an NMI in a designated metrological area
DIN	Deutsches Institut für Normung e. V., the German standardisation body
DMP	Data Management Plan. It defines the policy and structure for storing the data generated during the project
EA	European co-operation for Accreditation
EU	European Union
EURAMET	European Collaboration on Metrology – a legal entity and the collaborative organisation for the NMIs of Europe
FSM	Force Standard Machine
GULMET	Regional Metrology Organisation of Gulf Countries: Gulf Association for Metrology
GULFMET TC-M	GULFMET Technical Committee for Mass and Related Quantities
HSM	Hardness Standard Machine
HTM	Hardness Testing Machine
IMEKO	International Measurement Confederation
IMEKO TC 5	International Measurement Confederation: Technical Committee of Hardness Testing
IMS	Indentation Measurement System
ISO	International Organisation for Standardisation
KCDB	Key Comparison Data Base. An open access database recording results and details of key comparisons worldwide and recognised CMCs of NMIs and DIs
NA	Numerical Aperture
NMI	National Metrology Institute – a national laboratory holding, disseminating and researching physical standards
RMO	Regional Metrology Organisation
SI	Système International d'Unités (International System of Units)
TURKAK	Turkish Accreditation Agency

Contents

Section A: Key data	4
A1 Project data summary	4
A2 Financial summary	5
A3 Work packages summary	5
Section B: Overview of the research	6
B1 Summary of the project	6
B2 Excellence	9
B2.a Overview of the objectives	9
B2.b List of deliverables	9
B2.c Need for the project	10
B2.d Progress beyond the state of the art	11
B2.e Gender dimension	13
B2.f Open science	13
B2.g Research data management and management of other research outputs	15
B3 Potential outcomes and impact from the project	16
B3.a Projected outcomes for industrial and other user communities	16
B3.b Projected outcomes for the metrological and scientific communities	16
B3.c Projected outcomes for relevant standards	17
B3.d Projected wider impact of the project	17
B3.e Summary of the project's impact pathway	18
B4 The quality and efficiency of the implementation	19
B4.a Overview of the consortium	19
Section C: Detailed project plans by work package	22
C1 WP1: Development of definition and measurement methodology for hardness indentations of Brinell, Vickers and Knoop	22
C1.a Task 1.1: Definition of indentation boundaries in Brinell, Vickers, and Knoop	22
C1.b Task 1.2: Influence of the focal point on optical indentation size measurements	23
C1.c Task 1.3: Influence of illuminance on optical indentation size measurements	23
C2 WP2: Development of transfer standards for Brinell, Vickers and Knoop indentations	24
C2.a Task 2.1: Selection of materials to be used for indentation transfer standards	24
C2.b Task 2.2: Optimisation of the design of the indentation transfer standards	25
C2.c Task 2.3: Determination of hardness scales and levels for indentation transfer standards	25
C2.d Task 2.4: Production of the indentation transfer standards	26
C3 WP3: Traceability chain from NMI to user level	27
C3.a Task 3.1: Review of hardness testing machines, standards, and applications	27
C3.b Task 3.2: Investigation of influencing factors in Brinell, Vickers and Knoop hardness tests	28
C3.c Task 3.3: Advanced uncertainty models for Brinell, Vickers and Knoop hardness tests	29
C3.d Task 3.4: Development of a traceability chain methodology for Brinell, Vickers and Knoop hardness tests	30
C4 WP4: Automatisisation of indentation measurements	31
C4.a Task 4.1: Review the current indentation measurement systems and software	32
C4.b Task 4.2: Modifying the present system at TUBITAK to enable marking of indentation locations	33
C4.c Task 4.3: Improvement of the TUBITAK system software	33
C4.d Task 4.4: Analysis of indentations suitable for automatic measurements	34
C5 WP5: Creating impact	35
C5.a Task 5.1: Dissemination and communication	35
C5.b Task 5.2: Exploitation and uptake	37
C6 WP6: Management and coordination	38
C6.a Task 6.1: Project management	38
C6.b Task 6.2: Project meetings	39
C6.c Task 6.3: Project reporting	39
C7 Gantt chart	40
Section D: Risk and risk mitigation	46
D1 Scientific/technical risks	46
D2 Management risks	47
D3 Ethics	48
Section E: References	50

Section A: Key data

A1 Project data summary

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Participant details:

a. Participants

no.	Participant Type	Short Name	Organisation legal full name	Country
1	Internal Beneficiary	TUBITAK	Türkiye Bilimsel ve Teknolojik Arastırma Kurumu	Türkiye
2	Internal Beneficiary	BEV-PTP	Physikalisch-Technischer Pruefdienst des Bundesamt fuer Eich- und Vermessungswesen	Austria
3	Internal Beneficiary	CMI	Cesky Metrologický Institut	Czechia
4	Internal Beneficiary	GUM	Central Office of Measures	Poland
5	Internal Beneficiary	IMBiH	Institut za mjeriteljstvo Bosne i Hercegovine	Bosnia and Herzegovina
6	Internal Beneficiary	INRIM	Istituto Nazionale di Ricerca Metrologica	Italy
7	Internal Beneficiary	PTB	Physikalisch-Technische Bundesanstalt	Germany
8	External Beneficiary	INM	I.P. Institutul Național de Metrologie	Moldova, Republic of
9	External Beneficiary	POLITO	Politecnico di Torino	Italy
10	External Beneficiary	UNZE	Javna ustanova Univerzitet u Zenici Universitas Studiorum Zenicaensis	Bosnia and Herzegovina
11	External Beneficiary	ZAG	Zavod za gradbeništvo Slovenije / Slovenian National Building and Civil Engineering Institute	Slovenia
12	Unfunded Beneficiary	EASYDUR	EASYDUR SRL	Italy
13	Unfunded Beneficiary	LTF	LTF S.p.A.	Italy
14	Unfunded Beneficiary	MPA NRW	Materialprüfungsamt Nordrhein-Westfalen	Germany

A2 Financial summary

	Internal Beneficiaries	External Beneficiaries	Unfunded Beneficiaries	Associated Partners	Total
Labour (€)	275 700.00	60 300.00	37 800.00		373 800.00
Subcontracts (€)	12 500.00				12 500.00
T&S (€)	43 250.00	23 000.00	15 000.00		81 250.00
Equipment (€)					
Other Goods, Works and Services (€)	42 500.00	7 500.00	3 000.00		53 000.00
Internally Invoiced Goods and Services (€)					
Indirect (€)	90 362.50	22 700.00	13 950.00		127 012.50
Total costs (€)	464 312.50	113 500.00	69 750.00		647 562.50
Costs as % of Total costs	72 %	18 %	11 %	0 %	
Total Eligible Costs (€)	464 312.50	113 500.00	69 750.00		647 562.50
EU contribution (€)	464 312.50	113 500.00			577 812.50
EU contribution as % of total EU contribution	80 %	20 %	0 %	0 %	
Months	73.0	19.0	6.3		98.3

TUBITAK will subcontract work in Task 4.3. This subcontracted work is required as a specialist is needed for the development of IMS software according to the project's specifications. The total cost of the subcontract is expected to be €12 500. This work will be subcontracted as the consortium does not have the necessary skills / expertise for such work.

A3 Work packages summary

WP No	Work Package Title	Active Participants (WP leader in bold)	Months
WP1	Development of definition and measurement methodology for hardness indentations of Brinell, Vickers and Knoop	PTB , TUBITAK, INRIM, POLITO, ZAG, MPA NRW	11.7
WP2	Development of transfer standards for Brinell, Vickers and Knoop indentations	PTB , TUBITAK, BEV-PTP, CMI, INRIM, UNZE, GUM	19.4
WP3	Traceability chain from NMI to user level	INRIM , TUBITAK, CMI, IMBIH, PTB, POLITO, UNZE, ZAG, EASYDUR, LTF	18.9
WP4	Automatisation of indentation measurements	TUBITAK , BEV-PTP, INRIM, PTB, INM	14.1
WP5	Creating impact	ZAG , TUBITAK, BEV-PTP, CMI, IMBIH, INRIM, PTB, INM, POLITO, UNZE, EASYDUR, LTF, MPA NRW, GUM	19.6
WP6	Management and coordination	TUBITAK , BEV-PTP, CMI, IMBIH, INRIM, PTB, INM, POLITO, UNZE, ZAG, EASYDUR, LTF, MPA NRW, GUM	14.6
Total months			98.3

The information in tables A2 and A3 reflect the estimates of resources as of the start of project in September 2023. The tables will not necessarily be updated during the course of the project.

Section B: Overview of the research

B1 Summary of the project

Overview

Hardness is an important material property determined by measuring an indentation size realised on the tested material surface in the Brinell, Vickers and Knoop (BVK) scales. This project will investigate the indentation measurement phenomena to provide a better defined, more consistent, unified and reliable measurement and traceability methodology to overcome the inconsistency between NMIs and lower levels of indentation (hardness) measurements as well as between NMIs. The project outcomes will be used in the next generation of hardness definitions, instrumentation and standardisation to improve the accuracy of material testing in all engineering fields, including aerospace, automotive, health, industry, and research and development.

Need

Material properties are a crucial factor in construction, design and engineering. Knowledge of the critical mechanical properties of materials used in production and construction is essential to prevent health and safety hazards, control the costs of production, and create better designed, healthy and consumer friendly products. This knowledge is obtained through the establishment of accurate, reliable and consistent reference standards for material properties at NMIs and a link to end-user measurements for the lowest level test in a workshop through a continuous traceability chain. Brinell, Vickers and Knoop testing methods are three of the most important ones used for testing the hardness of metallic materials. Hardness is determined in these scales by measurement of the size of an indentation made with an implement of known geometry and properties on the surface of the material with a known force. There are many types of indentation measurement systems and methodologies. The indentation size measurement is completely dependent on the image magnification and processing instrument, the operator, and software if any. There are significant hardness measurement inconsistencies among NMIs, as well as between NMI and testing laboratories due to a lack of a well-defined methodology and specifications for the variety of indentation measurement instruments used. For example, optical microscopes with different numerical apertures (NAs) and software give different measurement results. This issue was demonstrated in 2005 at the NMI level in the CIPM Brinell Key Comparison, CCM.H-K2. At the lower levels of the measurement chain, the problem is much more evident. Traceable propagation of the reference value from NMIs to testing laboratories is not achieved due to the variety of simple devices used for testing without controlling all parameters. This project aims to overcome these critical problems by establishing the necessary methodologies and references for consistent, unified, reliable and traceable hardness indentation measurements from the NMI level to testing laboratories.

Objectives

The overall objective of this project is to establish a traceable, consistent, and reliable indentation measurement methodology, considering the influence of measurement parameters and instrument specifications, for Brinell, Vickers and Knoop hardness scales. The specific objectives are as follows.

1. To establish universal definitions for indentation boundaries for Brinell, Vickers and Knoop hardness scales and to determine the required measurement parameters to be used with the highest accuracy and most commonly used instruments at the NMI level (scanning and optical) for the detection and measurement of indentation dimensions in line with relevant ISO standards.
2. To develop and validate stable reference indentation artefacts representative of the full Brinell, Vickers and Knoop hardness scales for propagation of traceability from NMI level to end users and to ensure inter-NMI comparability and consistency of indentation measurements.
3. To develop a methodology to ensure robust traceability for Brinell, Vickers and Knoop hardness measurements, from NMI to user level, through determination of the effect of influence factors on measurement results, and the development and validation of uncertainty models considering the most commonly used instrumentation at all levels and the applicable documentary standards. Additionally, to produce recommended technical specifications for manufacturers of hardness calibration and testing instruments.
4. To develop the software of an existing indentation measurement system (IMS) to automatise the determination of indentation locations and their measurements where possible and to determine the types of indentations that are suitable for such measurements and classify them.

5. To facilitate the take up and long term operation of the capabilities, technology and measurement infrastructure developed in the project by the measurement supply chain (NMIs/DIs, calibration and testing laboratories), standards developing organisations (ISO), and end users (e.g. industry, regulators, manufacturers). The approach should be discussed within the consortium and with other EURAMET NMIs/DIs e.g. EURAMET TCs or EMNs, to ensure that a coordinated and optimised approach to the development of traceability in this field is developed for Europe as a whole.

Progress beyond the state of the art and results

Development of methodology for Brinell-Vickers-Knoop indentation measurement:

Indentation measurement is generally realised with optical measurement devices, such as an optical microscope, with which two perpendicular diameters for Brinell, two diagonal length of Vickers and one diagonal length of Knoop indentations are measured. After inconsistent results were found in indentation measurements performed in the CIPM Brinell Key Comparison, CCM.H-K2, in 2005, the reason was found to be a significant effect caused by the use of lenses with different numerical apertures (NAs). It is now advised to use NAs equal to or smaller than 0.4 and measurement traceability is established via a certified stage micrometer (line scale). Yet, there is still no firm definition for the borders of indentations in all three scales (BVK) needed to determine the start and end points of the dimensional indentation measurements, which continues to be a source of measurement inconsistencies.

This project will establish a definition for the border of the circular shaped indents in the Brinell scale for different sizes and shapes (sink-in vs pile-up), for the corners of the square shaped indent in Vickers and the equilateral indent in the Knoop scale for different shapes (hard and soft). Then, specification will be determined for recognising the borders and corners the indentations. At the conclusion of this project, end-users will have precise definitions of the measurands in the Brinell, Vickers and Knoop scales, and a standardised method for identifying their boundaries.

Development Brinell-Vickers-Knoop reference indentations:

The traceability in indentation measurement is established by calibration of the IMS via a certified stage micrometer and a reference indentation on a hardness reference blocks, but these indentations are certified only by the NMI or calibration laboratory with their own IMS. The traceability in 2D (via stage micrometer) is easy to manage because the marking on the stage micrometer is more obvious and the imaging system and operator effect is much less. However, in actual indentation measurements, results vary according to system performance and imaging parameters due to the 3D characteristic of the indentations.

This project will design and develop a set of long-term stable reference indentations for representative scales and hardness levels to serve as transfer standards which will ensure accurate dissemination of the scales and traceability of measurement results.

Development of traceability chain from NMI to user level:

Hardness testing machines are produced without regard to considerations of traceability or correlation between the highest measurement systems and the simpler ones used in industry. There is lack of understanding of relevant definitions and traceability requirements in the applicable ISO and ASTM hardness standards which causes very significant deviations in implementation.

This work will develop a methodology to unify propagation of traceable and consistent indentation measurements from NMI and DI level to calibration and testing laboratories. Furthermore, recommendations will be drafted for producers of hardness calibration and testing machines to ensure the compatibility of the machines they produce with higher level systems and the consistency of measurement results throughout the measurement chain.

Automatisation of indentation measurements:

Indentation measurement systems at NMIs and DIs operate either manually or semi-automatically. Semi-automatic measurement enables measurement of the size of an indent in one direction after which the indentation must be manually rotated by 90° to make the second diameter or diagonal length measurement. This can be done only for one indentation although there have been trials to measure more than one indentation by specifying the location manually by the operator before their realisation. In entirely manual measurement systems, the border of indentations is decided by the operator which means there may be significant inconsistency between measurements performed by different operators.

In this project, the indentation measurement system present at TUBITAK will be revised to improve its software and mechanical components to demonstrate full automatisisation of indentation measurements. After the revisions, the locations of indents will be selected through the software and marked on a hardness block by the indentation measurement system (IMS) automatically, the indents will then be realised on the block by

primary standard machines, and then the block will be placed in the IMS in its first position where the indentation locations were marked and saved by the software and indentation measurements will be realised automatically. This will be done for indentations for which edge detection with image processing are possible. Differently sized and shaped indentations will be analysed to determine the characteristics of those suitable for automatic measurements.

Outcomes and Impact

Outcomes for industrial and other user communities:

This project aims to improve the reliability of hardness tests at the industrial level by providing a better defined hardness (indentation) measurement methodology and sets of reference indentations agreed upon by the NMIs to provide traceability in BVK hardness scales. More importantly a methodology for traceability from the NMI level to the end user will be developed and based on this a set of recommendations for hardness testing instrument producers will be produced and disseminated. Uptake of the recommendations will enable a new generation of instrumentation for BVK hardness tests that produce results that are more consistent with the higher level machines. On a wider scale, more accurate, reliable and consistent hardness measurements will benefit all industries in which such measurements are central to producing higher quality, durable and safer products.

Outcomes for the metrology and scientific communities:

The problem at the core of this project is the source of the traceability in hardness measurements, which is the definition of the indentation and the methodology for its measurement. The first outcome of the project will rely on a scientific approach to indentation phenomena to arrive at a universally applicable and clear definition of its borders, and accordingly, a uniform method for its measurement. This will enable more accurate and consistent measurement results, and consequently, improved hardness calibration and measurement capabilities (CMCs) at NMIs and DIs and a significant advancement in hardness metrology. Automatisations of hardness measurements, which will be demonstrated at an NMI in this project, also brings the prospect of faster and more cost-efficient measurements which are less affected by operator related errors.

Outcomes for relevant standards:

The main outputs of the project, including BVK indentation definitions, uncertainty models and methodology for establishment of traceability for BVK hardness measurements will be used as input to improve ISO 6506-1, ISO 6506-2, and ISO 6506-3 for Brinell, ISO 6507-1, ISO 6507-2, and ISO 6507-3 for Vickers, and ISO 4545-1, ISO 4545-2, and ISO 4545-3 for Knoop hardness tests, which currently have significant deficiencies. ASTM standards ASTM E92, ASTM E384 and ASTM E10 may also be improved as a consequence. Standardisation is envisioned to be one of the main conduits for widespread uptake of project outputs by industry and other end-users.

Longer-term economic, social and environmental impacts:

More consistent, reliable and traceable measurements of material properties will enable better, safer product design and manufacturing at a lower cost. Hardness tests are a widely used material testing method to obtain preliminary information about abrasion and strength properties of a material easily and quickly. The durability of a selected material is critical to the proper, predictable and safe functioning of manufactured products and components and the safety of the built environment.

Europe is one of the largest producers and exporters of highly engineered products, including aircraft, automobiles and other vehicles, defence systems, durable households' goods, etc, for which accurate and reliable knowledge of constituent materials is key to quality, and by extension, product marketability and demand. Having access to reliable and traceable hardness measurements will thus benefit European industry and its competitive position on global markets. More directly, the uptake of the recommended technical specifications for hardness testing machines by manufacturers will ensure that European producers will have an edge over their rivals in this field.

Failures of material in wastewater and waste disposal infrastructure and in systems for storage and transportation of toxic and hazardous substances are an instance of a source of environmental damage that could be ameliorated through better measurement of material properties so that their critical failure points are known with greater accuracy and reliability. Inconsistent and incomparable measurements, as is currently prevalent in hardness testing, introduce an unacceptably high level of risk of failure in systems needed for the safe handling of toxic materials and materials. With better testing of hardness, and therefore of material durability, the infrastructure, systems and products that are relied upon to protect the environment can be expected to be more suitable for purpose, longer lasting and less likely to unpredictably fail.

Finally, safer and more reliable products, a reduction in the likelihood of hazards and accidents caused by product failure, and an appreciable improvement in the quality of the built environment will serve to enhance societal health and wellbeing.

B2 Excellence

B2.a Overview of the objectives

The overall objective of this project is to establish a traceable, consistent, and reliable indentation measurement methodology, considering the influence of measurement parameters and instrument specifications, for Brinell, Vickers and Knoop hardness scales. The specific objectives are as follows.

1. To establish universal definitions for indentation boundaries for Brinell, Vickers and Knoop hardness scales and to determine the required measurement parameters to be used with the highest accuracy and most commonly used instruments at the NMI level (scanning and optical) for the detection and measurement of indentation dimensions in line with relevant ISO standards. (WP1)
2. To develop and validate stable reference indentation artefacts representative of the full Brinell, Vickers and Knoop hardness scales for propagation of traceability from NMI level to end users and to ensure inter-NMI comparability and consistency of indentation measurements. (WP2)
3. To develop a methodology to ensure robust traceability for Brinell, Vickers and Knoop hardness measurements, from NMI to user level, through determination of the effect of influence factors on measurement results, and the development and validation of uncertainty models considering the most commonly used instrumentation at all levels and the applicable documentary standards. Additionally, to produce recommended technical specifications for manufacturers of hardness calibration and testing instruments. (WP3)
4. To develop the software of an existing IMS to automatise the determination of indentation locations and their measurements where possible and to determine the types of indentations that are suitable for such measurements and classify them. (WP4)
5. To facilitate the take up and long term operation of the capabilities, technology and measurement infrastructure developed in the project by the measurement supply chain (NMIs/DIs, calibration and testing laboratories), standards developing organisations (ISO), and end users (e.g. industry, regulators, manufacturers). The approach should be discussed within the consortium and with other EURAMET NMIs/DIs e.g. EURAMET TCs or EMNs, to ensure that a coordinated and optimised approach to the development of traceability in this field is developed for Europe as a whole. (WP5)

B2.b List of deliverables

Relevant objective (Activity delivering the deliverable)	Deliverable number	Deliverable description	Deliverable type	Participants (Lead in bold)	Delivery date
1 (A1.1.7)	D1	Report on the development of universal definitions for Brinell, Vickers and Knoop (BVK) hardness indentation boundaries	Report	PTB , TUBITAK, INRIM, POLITO, ZAG, MPA NRW	Sep 2025 (M25)
1 (A1.3.7)	D2	Report on required measurement parameters to be used with the highest accuracy and most commonly used instruments at the NMI level (scanning and optical) for the detection and measurement of indentation dimensions in line with relevant ISO standards	Report	PTB , TUBITAK, INRIM, POLITO, ZAG, MPA NRW	Feb 2025 (M18)
2 (A2.4.7)	D3	Report on the development and validation of reference Brinell, Vickers and Knoop indentations	Report	PTB , INRIM, TUBITAK, GUM,	Feb 2025 (M18)

		to be used as transfer standards		BEV-PTP, CMI, UNZE	
3 (A3.3.4)	D4	Report on the validation of advanced uncertainty models developed for Brinell, Vickers and Knoop hardness tests	Report	INRIM , TUBITAK, CMI, PTB, IMBIH, POLITO, UNZE, ZAG	Feb 2026 (M30)
3 (A3.4.2)	D5	Guidelines for the establishment of traceability in Brinell, Vickers and Knoop hardness indentation measurements from NMI to end user level	Guidelines	INRIM , TUBITAK, CMI, PTB, IMBIH, POLITO, UNZE, ZAG	Jul 2026 (M35)
3 (A3.4.3)	D6	List of recommendations for hardness instrument producers including technical specifications	Report	INRIM , TUBITAK, PTB, EASYDUR, LTF	Jul 2026 (M35)
4 (A4.3.6)	D7	User manual for the software/algorithm developed for automated Brinell, Vickers and Knoop indentation measurements	User Manual	TUBITAK , INRIM, PTB	Apr 2026 (M32)
4 (A4.4.4)	D8	Report on the analysis of the results of automatic Brinell, Vickers and Knoop indentation measurements with specifications for the types of indentations suitable for automatic measurements	Report	TUBITAK , INRIM, PTB	Aug 2026 (M36)
(n/a)	D9	Evidence of contributions to or influence on new or improved international guides, recommendations and standards with a specific focus on the following guides and committees: CIPM/CCM WGH, ISO TC 164 SC3, EURAMET TCM WG Hardness, Hardness instrument producers (primary, secondary, testing), Accreditation bodies. Examples of early uptake of project outputs by end-users. Updated dissemination, communication and exploitation plan.	Reporting documents	TUBITAK , all participants	Aug 2026 (M36) + 60 days
n/a	D10	Delivery of all technical and financial reporting documents as required by EURAMET	Reporting documents	TUBITAK , all participants	Aug 2026 (M36) + 60 days

B2.c Need for the project

The measurement of hardness, a critical property of materials that affects how and where they can be safely used in a vast array of industrial products, construction, and infrastructure, is complicated by a lack of consistency and comparability, both between NMIs and between NMIs, secondary laboratories and end-users. An interlaboratory comparison made of Brinell hardness scale measurements in 2005 at the NMI level revealed significant deviations in measurement results between the participants that was found to be due to the variations in the numerical aperture (NAs) used in the imaging system. In hardness block calibrations, measurement results that are frequently inconsistent with the nominal values of the blocks highlight the lack

of standardised measurement methodology and instrumentation, although all instruments used are compliant with relevant ISO hardness standards from a length measurement point of view. The same situation occurs in testing machines; different types of testing machines may differ in their measurements of the same reference blocks although they comply with the relevant hardness standards in terms of each component constituting the hardness scale such as force application, indenter used, indentation measurement system and measurement cycle. This is largely the result of discrepancies in the indentation measurement systems related to:

- lack of a commonly accepted definition of the measurand (indentation) and uniformity in the methodology used,
- lack of hardness measurement traceability between NMIs and from the NMI level to the end user,
- lack of a traceability chain that extends to the producers of instrumentation, and,
- critical dependence on the operator's subjective decision on the precise location of the indentation borders.

These are the important parts of an IMS that are currently not well defined, characterised and unified.

Diameter measurements of Brinell indentations in sink-in, pile-up, isotropic or non-isotropic materials give different results with different imaging and measurement systems because there is neither a physical definition for nor a common agreement on the location of the border of indentation. It is important to objectively define or at least agree on the boundaries of the indentation and its diameter as a starting point of the traceability that can be propagated from the highest level of measurement to the end-user. Consistent measurement demands that NMIs and DIs deliver a standardised measurand to the calibration and testing laboratories which can then also be used by industry. It is also necessary to adequately define and agree upon the conditions under which measurements shall be performed to supplement unified and standardised Brinell indentations in the world of metrology. The same situation applies for Vickers indentations as well as Knoop ones, this time to define the corners of indentation which are the end points of the diagonal length that are measured for determination of hardness values of material.

Once a reference definition is constituted it needs to be propagated along the measurement chain. This makes it necessary to develop long-term stable and high quality real/dummy reference indentations to be used as transfer standards for propagation of the indentation measurement methodology as well as assurance of the traceability and consistency of measurement results. The proposed set of reference indentations will ensure the traceability of real indentation measurements in three dimensions rather than two dimensions as is currently possible. The set of indentations will be produced for all hardness scales and levels used in industry giving rise to the need to settle discrepancies between different hardness scales, which the project aims to address as well.

At the testing laboratory level, the problem in achieving consistent and traceable hardness measurements is much more obvious due to the variety of simple devices used for testing without controlling all parameters. Different magnification, imaging and measurement devices result in varying determinations of the border of indentations and thus indentation sizes, which ultimately results in inconsistent hardness measurements. This situation introduces the necessity to investigate the link between the industrial testing devices and the reference devices present at the NMIs or DIs. Determining the correlation between the reference systems and the testing systems is one of the most important parts of this project as it can guide the construction and design of the next generation of hardness measuring machines. The list of recommendations that will be provided to hardness instrument producers will enable consistent and comparable measurements and will change the future development of these instruments.

Finally, after defining the measurand (indentation) and ensuring its propagation to the lower level of measurement stages, another problem confronted is not related to either the definition of the measurand or to the instrument itself, but to the operator, the person who performs the measurement. Here another parameter that brings about deviations in measurements are the decisions taken by the operators about the location of the indentation borders. Automatisations of measurements is seen as the means to overcome this issue, while also resulting in labour and cost efficiencies.

B2.d Progress beyond the state of the art

Development of methodology for Brinell-Vickers-Knoop indentation measurement:

Indentation measurement is generally realised with optical measurement devices such as optical microscope with which two perpendicular diameters for Brinell, two diagonal lengths of Vickers and one diagonal length of Knoop indentations are measured. There is currently neither a standardised methodology nor a technical specification for the optical or any other scanning devices that are used to measure the indentations. Indeed,

there is no universally accepted method to determine the location of the indentation, i.e., the start and end points of the diameter of Brinell indents, the location of the corner points for the square shaped Vickers indents and equilateral Knoop indents. Besides the definition of the indentation, it is also difficult to be certain about the reliability of the instrument to be used for measurement. Even in optical measurement devices, it can be difficult to decide on the magnification and imaging devices since different devices, with varying technical specifications for magnification and imaging, give different measurement results. For instance, there were significant inconsistencies between the measurement results of the NMIs in the CIPM Brinell Key Comparison, CCM.H-K2, in 2005 [12] although traceability in terms of length was assured before the comparison at NMIs level. An investigation found that the numerical aperture (NA) of the lens had a significant effect on the indentation measurement in Brinell hardness especially. It is now advised to use $NA \geq 0.4$ and the measurements are continued with traceability provided via a certified line scales (stage micrometers) without a definition for the border of indentations in all three scales. For routine measurements and calibration of hardness reference blocks, $NA \geq 0.4$ is used so as to conform to the “measurement methodology agreed upon”. This brings many difficulties in constituting a reference value and propagating it to the hardness measurement devices used in Europe. The same situation applies to Vickers and Knoop indents, i.e. it is difficult to determine the precise location of the corners in hard materials, the decision being reliant on operator experience.

In this project, the indentation phenomena will be investigated for different materials (steel, aluminium, brass, copper etc.) for different scales (different indenters such 1 mm, 2.5 mm, 5 mm and 10 mm diameter Tungsten carbide balls, square based pyramid and rhombic shaped diamond indenters, force values ranging from some 10 gf – 50 gf to maximum 3000 kgf), and different indentation measurement systems featuring optical imaging equipment such as optical microscopes and scanning probe microscopy devices. The most common and best definable indentations will be used to create precise definitions for the location of the indentation borders. The conditions under which the indentation is imaged and magnified will be defined, and as a result a standardised methodology will be constituted for indentation measurements. The methodology will include specifications for determining the boundaries (i.e. end points for length measurements) of indentations and for instrument settings. The standardised methodology will establish the source for the traceability of Brinell, Vickers and Knoop indentation measurements, which does not currently exist.

Development Brinell-Vickers-Knoop reference indentation:

The traceability of indentation measurements is currently underpinned by calibration of the IMSs via a certified stage micrometer, and a reference indentation on the hardness reference blocks certified only by the NMI or calibration laboratory using their own IMS. The traceability in 2D (via stage micrometer) is easy to manage because the border of the stage micrometer line is more obvious and the effect of the imaging system and operator (as well as the software, if any) effect is much less. However, in actual indentation measurements, results vary according to system performance and imaging parameters due to the 3D characteristic of the indentations.

In this project, a set of real indentations realised on specially produced hardness reference blocks will be produced to be used to confirm the consistency of the measurement standards of the NMIs participating in the project. In order to produce the reference indentations, different materials will be studied to serve as blocks upon which indentations of different sizes and types will be created with primary hardness standard machines and measured by participant NMIs. Once a standardised methodology has been established through the work performed in WP1, the sets of reference indentations will be provided to calibration laboratories and hardness instrument producers to be used for assurance of measurements according to the aforementioned methodology and to establish traceability from the NMI level to the user level.

Development of traceability chain from NMI to user level:

Hardness testing machines, simpler in construction than the systems used at NMIs, are not produced with traceability to primary standards, nor is the correlation of their measurements to those obtained from NMI systems known. There is lack of understanding of relevant definition and traceability requirements in the applicable ISO hardness standards (as well as those of ASTM) and these frequently cause very significant deviations in implementation. This means that when the secondary and testing machines are designed and developed, the discrepancies in measurement results due to instrument specifications are not taken into consideration. Traceability is achieved by certified length standards and while this may work for 2 dimensional measurements, traceability for 3 dimensional indentation measurements is not assured. The calibration of hardness testing machines is thus valid for some scales and/or hardness levels but not for others.

In this project, indentation measurement instruments produced in Europe will be analysed and the instrument, measurement methodology and the software, where possible, will be studied. A correlation between the primary measurement standards (based on the measurement results of the reference indentations produced in WP2) and the types of testing machines most widely used in Europe will be

established. Based on this, recommendations will be prepared for manufacturers of instrumentation, both general and specific to each of the testing machines analysed, to ensure traceable and consistent indentation measurements from the NMI level systems to calibration and testing machines in industry.

One of the main focuses will be to produce guidelines for the establishment of traceability in Brinell, Vickers and Knoop hardness test measurements from NMI to end user level. This will be aimed at improving hardness metrology in emerging EURAMET countries by developing metrological capability of the participating NMIs and establishing the required traceability chain. It will also promote and improve the best practice for hardness measurements.

Automatisation of indentation measurements:

Indentation measurements at the NMI level are either manual or semi-automatic. In the semi-automatic case automatisation enables measurement in one direction of the Brinell, Vickers and Knoop indents, necessitating manual rotation of the indentation by 90° to make the second diameter or diagonal length measurement, necessitating the precise alignment of the diameter or the diagonal length to be measured in the same direction as the first measured one, which is a source of error. There are few trials to start the measurement of one indentation in 2 directions and the possibility to measure more than one indentation by specifying the location of n indents manually by the operator after their realisation. In the manual measurement systems, the border of indentations is decided by the operator which can result in significant inconsistencies between different operator measurements.

In this project, revisions will be made on a four motorised stage IMS present at the TUBITAK Hardness Laboratory. First, a mechanical part to make a circular mark on the surface of the block will be designed and adopted to the present system. With this part's help, the operator will place the hardness reference block on the plate of the IMS, and the region where the indentation measurements will be made will be adjusted by the operator and recorded by software to be developed in the project. The IMS system will then mark the indentation location automatically (with the help of the four motorised stages). After the marking process is finished, the indentations will be realised in the traditional way with the hardness standard machines and the block will be placed as it was for marking the indents' places. The IMS will make measurements of the indentation within those circular marks with the help of image processing and edge detection in accordance with the definition of indentation. Various indentations will be investigated and the ones suitable for this process (in terms of hardness scales and levels) will be selected and this automatisation will be used for such indents.

B2.e Gender dimension

The actual research itself in this project does not have a gender dimension as its focus is on measurement science of physical parameters, specifically, improving the measurement of hardness as a property of material. The outcomes of the research may eventually have impact on how and which particular materials are used in products, and by extension on product design, manufacturing and usage, which may have differing implications for different genders; however these considerations are beyond the scope of this project.

In terms of its dissemination activities, the project will communicate in a manner that highlights to all genders the benefits of improved metrological approaches for more reliable hardness calibrations.

B2.f Open science

Open science is an approach based on open cooperative work and systematic sharing of knowledge and tools as early and widely as possible in the process. The project will include open science practices as they have the potential to increase the quality and efficiency of research and accelerate the advancement of knowledge and innovation by sharing results, thus making them more reusable and improving their reproducibility.

The project will adopt open science practices as an integral part of the project. The project will follow open science practices as per Regulation (EU) 2021/695 establishing Horizon Europe – the Framework Programme for Research and Innovation, laying down its rules for participation and dissemination

(i) Early and open sharing of research:

Pre-prints of the project manuscript(s) will be submitted to a suitable open access platform, in order to accelerate the dissemination of results, ahead of publishing. Preprints will be submitted by the project to widely used and accessible repositories, selecting those that are most relevant to their stakeholders. Target pre-print archives will include: arXiv and Preprints.

(ii) Research output management:

The project will manage the digital research data generated in the project responsibly, in line with the FAIR principles and according to the requirements in the Grant Agreement. This will include:

- Establishing a data management plan (DMP) and regularly updating it.
- As soon as possible and within the deadlines set out in the DMP, deposit the data in a trusted repository (possibly listed in the Registry of Research Data Repositories).
- As soon as possible and within the deadlines set out in the DMP, ensure open access -via the repository- to the deposited data, under the latest available version of the Creative Commons Attribution International Public License (CC BY) or a licence with equivalent rights.
- Providing information via the repository about any research output or any other tools and instruments needed to re-use or validate the data.
- Ensuring metadata of deposited data will be open under CC 0 or equivalent, in line with the FAIR principles and provided with information as per the conditions in the Grant Agreement

(iii) Measures to ensure reproducibility of research outputs:

The project will provide information about the research outputs/tools/instruments needed to validate the conclusions of scientific publications or to validate/re-use research data. TUBITAK, INRIM, PTB, IMBiH, CMI, GUM, BEV-PTP and INM are either accredited to or work in compliance with ISO/IEC17025. This enables the participants to demonstrate that they operate competently and generate valid results, thereby promoting confidence in their work.

The project will provide digital or physical access to the results needed to validate the conclusions of scientific publications.

The project will ask its Stakeholder Committee to provide input and feedback on the project's work plan to support the robustness and reproducibility of its methods and research outputs.

(iv) Providing open access to research outputs:

The project will undertake responsible management of research data in line with the FAIR principles and open access to research data under the principle 'as open as possible, as closed as necessary', as per the Grant Agreement. The project will provide open access to scientific publications under the conditions in the Grant Agreement:

- At the latest at the time of publication, a machine-readable electronic copy of the published version or the final peer-reviewed manuscript accepted for publication, will be deposited in a trusted repository for scientific publications.
- Immediate open access will be provided to the deposited publication via the repository, under the latest available version of CC BY or a licence with equivalent rights.
- Information will be given via the repository about any research output or any other tools and instruments needed to validate the conclusions of the scientific publication.
- Metadata of deposited publications will be open under a CC 0 or equivalent licence, in line with the FAIR principles and provided with information as per the conditions in the Grant Agreement

(v) Participation in open peer-review:

The project will consider Open Research Europe (ORE) as a possible target journal for the project's publications. ORE is an open peer-review journal recently set-up by the EC for use by their project's participants. It is a no-fee, open access, peer-reviewed publishing venue for EU-funded research, where reviewers' comments and recommendations and authors' feedback are open for others to view.

(vi) Involving all relevant knowledge actors in the co-creation of R&I agendas and contents:

Meetings and workshops with the Stakeholder Committee involving key stakeholders and end users will be held to gain input and feedback on the project's work plan, results and impact activities.

Further to this, the project will use social media and knowledge sharing channels (e.g. LinkedIn, Twitter) to involve stakeholders and gain their feedback on and input to the project.

Young researchers and Postdocs will also be invited to participate in training events and to join stakeholder workshops to provide feedback.

B2.g Research data management and management of other research outputs

It is intended that the project data and outputs are available for use as widely as possible within the stakeholder community and are readily accessible for any subsequent follow-on project. The project will ensure that all relevant research data are soundly managed so as to make it Findable, Accessible, Interoperable and Reusable (FAIR).

Types of research data

The data/research outputs generated will be from measurements, calibrations, designs and analysis. The project will collect the following types of data: images in JPEG, PNG and others appropriate formats, numerical data in CSV, JSON, XML and other appropriate formats, text description data in Markdown format. Research outputs will include new software, calibration methods, protocols, materials. The estimated overall size of the data/research outputs is expected to be in the range 200 GB – 1 TB, but extended as required.

Findability of data/research outputs

The data/research outputs (protocols, reports, software) will be findable as each will be identifiable with a commonly used persistent and unique identifier such as DOI, Git commit/tag, Handle, persistent and unique identifier. The data will provide information on the following: bibliographic information (author(s), dataset description, date of deposit, venue, possibly embargo); information on funding (European Partnership on Metrology funding, project name, acronym, and grant number); licensing terms. Where applicable, it will include persistent identifiers for related publications and research outputs. The data / research outputs will be deposited and published in trusted repositories located using the Registry of Research Data Repositories. Software will be stored, documented and version-controlled in a suitable repository, e.g. GitLab. Protocols produced by the project will be stored in a protocol exchange repository.

Accessibility of data/research outputs

All of the data that are needed to validate the results presented in scientific publications/research outputs will be made openly available as the default unless there is a specific reason not to publish it. Other data/research outputs may be made available. As some existing data/research outputs may have restricted access, access will only be provided after personal contact with the authors via the repository interface. The data/research outputs made available will also be considered: for example, pre-processed data will not be provided unless there is a clear reason for doing so.

Interoperability of data/research outputs

The datasets will use the trusted repository's basic metadata schema for administrative data, which is compliant with the recommended standards used by DataCite, BASE search, OpenAIRE. To guarantee the interoperability of project's data/research outputs, individual datasets will be described using affirmed discipline- specific vocabularies, standards, formats, and methodologies, such as: e.g. GUM, OBO, DICOM, NetCDF, HDF5, CityGML, INSPEC, ISO 9001. Due to the thorough usage of standard terminologies, no specific compatibility mappings will be necessary. If necessary, the project's datasets that will be deposited in the chosen trusted repository will include qualified references to other datasets from the same project and/or previous research.

Reusability of data/research outputs

The data/research outputs will either be licensed under the latest available version of the CC BY license or a license with equivalent rights as set out in the Grant Agreement. Users will be required to acknowledge the consortium and the source of the data in any resulting publications. Alternatively, the Public Domain Dedication License (CC 0) will be used. Templates will be documented with embedded instructions rather than with separate README files to maintain reusability by minimising the number of files required. The data are in a common format and can be read using widely available software (open source or commercial).

Any data published in open-access journals will be usable by third parties after the datasets have been deposited in a trusted repository. The data that does not relate to peer- reviewed publications will be made available for re- use on a case- by- case basis.

Curation and storage/preservation costs

The estimated curation and storage/preservation costs for making the data and research outputs FAIR are 100 € (personnel costs). These costs will be kept to a minimum by using i) suitable trusted repositories from the Registry of Research Data Repositories where no additional costs are associated with long-term preservation, and ii) by making only relevant data and outputs FAIR. The estimated curation and storage/preservation costs are included in the project's budget and will be claimed if compliant with the Grant Agreement's conditions.

Participant, person or team responsible for data management and quality assurance

This consortium will not establish a DAC. The coordinator, with support from the participants, will have overall responsibility for the management of data/research outputs and quality assurance. The coordinator will be responsible for coordinating updates to the data management plan and for deciding on a case-by-case basis which data/research outputs will be kept and for how long. The participant(s) that produced the data will be responsible for organising backup and storage, archiving, and for depositing the data/research outputs within the chosen repositories.

B3 Potential outcomes and impact from the project

B3.a Projected outcomes for industrial and other user communities

The project focuses on overcoming the inconsistency between the primary hardness standards, and more prominently between primary standards and hardness calibration or testing machines, due to lack of definition of the border of measured indentations as well as a standardised measurement methodology, and consequent inability to establish traceability from the primary machines to the testing machines used by end-users. All hardness measurement laboratories currently perform Brinell, Vickers and Knoop indentation measurements without an agreed-upon definition of indentation borders or measurement method. The applicable standard only requires the measurement of at least two perpendicular diameters of Brinell indentations and gives no further guidance. Given variations in the imaging equipment and indentation shapes, it is difficult to determine with precision where the indent starts and where it ends. Even when the lens used for magnification of the border of indent is changed, the measurement result also changes very sharply. It is similar for changes made in the imaging instrument, illumination, operator and the software used for measurement.

An important outcome of the project for industry and end-users will be development of the reference indentations to be used for checking the consistency between NMIs as well as establishment of traceability from primary standards to testing machines. The reference standards will provide industry with assurance of the reliability and traceability of their measurement.

A set of recommendations concerning specifications for hardness testing machines in Europe as well as others to ensure their compatibility with higher level standards will enhance the design and production of the coming generation of the hardness testing machines.

Besides accuracy and well defined traceability, automatisisation of the system is another important issue. In this case indentations that can be measured automatically will be investigated and the indentations that are suitable for automatic measurement will be determined and classified. Such indentations' specifications will be analysed and reported.

Project results will be disseminated to stakeholders in European industry, in particular, producers of hardness calibration and testing instruments.

B3.b Projected outcomes for the metrological and scientific communities

The project proposes a solution to the problem of inconsistent hardness measurements starting from its root, by establishing universal definitions for Brinell, Vickers and Knoop indentations, their borders and a methodology for their measurement for the most widely used IMSs. This will have an impact on the hardness metrology community and on materials science research.

In regard to hardness measurements and hardness standards from scientific point of view; a hardness standard is not a physical quantity, it is a method based quantity and thus the definition of the method and the measurands subject to consideration in hardness are very critical, as is the case in Brinell, Vickers and Knoop hardness indentation measurements. This constitutes the main core of the project; to define the indentations and their measurements., the starting point of the traceability to be propagated to calibration and test laboratories and in turn to all production lines. Building on this, new 3D indentation transfer standards, uncertainty models for BVK hardness tests, a methodology to ensure traceable measurements from primary standards to the end-user, and automatisisation of hardness measurements will all represent significant advances in hardness metrology. Their uptake and application by NMIs and DIs will ensure that world-leading services in hardness metrology will be available to the stakeholders that need them.

Developments and outcomes of the project will be shared with CIPM CCM Working Group for Hardness and EURAMET TC-M's newly established Working Group on Hardness, through which uptake of the knowledge and outputs by the metrology community will be facilitated.

B3.c Projected outcomes for relevant standards

The consortium will promote the results of the project within the standardisation community and will provide input into the standardisation process. Project participants will attend meetings of standardisation committee ISO/TC 164 Mechanical testing of metals and its SC 3 and WG 2, in order to communicate project developments, disseminate its outputs and contribute to the improvement of documentary standards. In particular, the definition of BVK hardness indentations, BVK hardness tests uncertainty models, and required technical specifications for BVK indentation measurement systems will be presented and proposals will be made for revisions to the series of ISO standards (ISO 6506, ISO 6507 and ISO 4545) applicable to BVK hardness tests. However, this process can be very lengthy and will extend beyond the duration of this project. Project developments will also be shared with a number of metrology committees including CIPM Consultative Committee for Mass (CCM) Working Group for Hardness, and the Technical Committees for Mass and Related Quantities of EURAMET, COOMET, and GULFMET, of which one or more project participants are members. Uptake of the knowledge, methods and other outputs generated by the project by the wider scientific and industrial communities will be assured through their incorporation into improved standards and guidance documents.

Standards Committee / Technical Committee / Working Group	Participants involved	Likely area of impact / activities undertaken by participants related to standard / committee
ISO/TC164 SC 3 and WG 2	PTB, INRIM, TUBITAK	ISO TC164 SC 3 (Hardness testing) and WG 2 (Measurement Uncertainty) meetings are held almost every year. PTB, INRIM and TUBITAK will take part in these meetings and share the project developments and outcomes. The outputs of this project, including the definition of BVK hardness indentations (D1), BVK hardness tests uncertainty models (D4), and required technical specifications for BVK indentation measurement systems (D2) will be disseminated to ISO TC 164 SC3 by PTB, INRIM and TUBITAK. Based on these outputs, a draft work item will be submitted for the revision of ISO 6506-1, 6506-2 and 6506-3 for Brinell, ISO 6507-1, 6507-2 and 6507-3 for Vickers and ISO 4545-1, 4545-2 and 4545-3 for Knoop hardness.
CIPM CCM WGH	PTB, INRIM, TUBITAK	The CIPM CCM Working Group on Hardness (WGH) meets nearly every year to discuss developments in hardness metrology. An important item on the CCM WGH's agenda is the redefinition of hardness scales. A better definition for Brinell, Vickers and Knoop hardness indentations and scientific results related to their measurements will be an important input to this discussion. Project will be shared by the PTB as the chair of the CCM WGH for the current period and by INRIM and TUBITAK as members.
EURAMET TC-M and WG Hardness	INRIM, PTB, TUBITAK	EURAMET TC-M WG Hardness was established to focus on hardness metrology activities in Europe. The progress and outcomes of this project will be placed on the WG meetings' agenda, and the developments in the project will be shared by INRIM, PTB and TUBITAK in every meeting during the course and after the completion of the project.
COOMET TC-M	PTB, TUBITAK	COOMET TC-M meets annually and encompasses the subject of hardness metrology. The progress and outcomes of this project will be shared by PTB and TUBITAK at the COOMET TC-M meetings.
GULFMET TC-M	TUBITAK	GULFMET was established in 2015 and became a fully recognised Regional Metrology Organisation in 2021. The Technical Committee on Mass encompasses hardness metrology and meets twice a year. The progress and outcomes of this project will be shared in all TC-M and GA meetings by TUBITAK which are an associate member and regular attendee of GULFMET meetings.

B3.d Projected wider impact of the project

Economic impact:

More consistent, reliable and traceable measurements of material properties will enable better, safer product design and manufacturing at a lower cost. Hardness tests are a widely used material testing method to obtain preliminary information about abrasion and strength properties of a material very easily and quickly.

The durability of selected material is critical to the proper, predictable and safe functioning of manufactured products and components and the safety of the built environment.

Europe is one of the largest producers and exporters of highly engineered products, including aircraft, automobiles and other vehicles, defence systems, durable households' goods, etc, for which accurate and reliable knowledge of constituent materials is key to quality, and by extension, product marketability and demand. Having access to reliable and traceable hardness measurements will thus benefit European industry and its competitive position on global markets. More directly, the uptake of the recommended technical specifications for hardness testing machines by manufacturers will ensure that European producers will have an edge over their rivals in this field.

Environmental impact:

Failures of material in wastewater and waste disposal infrastructure and systems for storage and transportation of toxic and hazardous substances are an instance of a source of environmental damage that could be ameliorated through better measurement of material properties, so that their critical failure points are known with greater accuracy and reliability. Inconsistent and incomparable measurements, as is currently prevalent in hardness testing, introduce an unacceptably high level of risk of failure within such systems that ensure toxic materials and materials are handled properly. With better controlled hardness, and therefore material durability, measurements; the infrastructure, systems and products that are relied upon to protect the environment can be expected to be longer lasting and less likely to unpredictably fail.

Social impact:

Consistent and traceable hardness measurements will result in safer and more reliable products, a reduction in the likelihood of hazards and accidents caused by product failure, and an appreciable improvement in the quality of the built environment, all of which will serve to enhance societal health and wellbeing.

B3.e Summary of the project's impact pathway

SPECIFIC NEEDS	EXPECTED RESULTS	DCE MEASURES
<i>What are the specific needs that triggered this project?</i> The field of materials testing needs consistent, reliable and traceable hardness measurements with lower uncertainties than currently available in order to boost the safety and quality of products in which the durability of constituent materials is a critical point of potential failure. At present, there are significant discrepancies between NMIs and between NMIs and industrial users in the measurement results of Brinell, Vickers and Knoop hardness measuring instruments even when applicable ISO and ASTM documentary standards are followed. The root of the problem lies with indentation measurements that are highly dependent on operator subjectivity and imaging system characteristics due to a lack of a precise definition for indentation borders and standardised methodology for indentation measurements.	<i>What do you expect to generate by the end of the project?</i> The project is expected to generate the following results: 1. Precise universal definitions for the Brinell, Vickers and Knoop indentations and specifications for determination indentation borders. 2. Clear and concrete specifications for hardness measuring instruments for more consistency between them in terms of recognising indentation boundaries. 3. Indentation measurement methodology for propagation of traceability from NMIs to testing labs 4. Indentation transfer standards to be used for assurance of the measurement standards and traceable calibration of testing devices. 5. An algorithm and software to enable automatic indentation measurements for the suitable instruments.	<i>What dissemination, communication and exploitation measures will you apply to the results?</i> Dissemination: 1. Communication with standardisation committees, EURAMET TC-M and CCM 2. At least 4 peer-reviewed open access papers scientific journals and 4 presentations at conferences. 3. Participation in CCM WGH activities related to definition of hardness scales. 4. Recommended instrument specifications for manufacturers of hardness standard, calibration and testing machines. 5. Reports and guideline documents to be made available via repositories Communication: 6. Project website created. 7. Stakeholder workshop held on project results. 8. Newsletters produced promoting the value of the results and outputs. These will be published on the project website and distributed to stakeholders. Exploitation:

		9. Adoption of standardised methodology for BVK indentation measurements by NMIs, testing laboratories 10. Submission of a proposal for revision of ISO series 6506, 6507 and 4545 standards 11. Production and distribution of indentation transfer standards 12. Automatisations of indentation measurements at participant NMIs where applicable.
TARGET GROUPS	OUTCOMES	IMPACTS
<i>Who will use or further up-take the results of the project? Who will benefit from the results of the project?</i> The target groups of this project are all entities dealing with hardness measurements in Brinell, Vickers and Knoop hardness scales as follows 1. Manufacturing companies, research institutes, universities and other entities performing hardness measurement 2. ISO, ASTM and national standardisation bodies 3. NMIs/DIs 4. CRM (hardness reference blocks) producers 5. CRM (hardness reference blocks) calibration laboratories 6. Hardness instrument (standard, calibration and testing machines) producers 7. All end users of CRMs traceable to the target groups mentioned above	<i>What change do you expect to see after successful dissemination and exploitation of project results to the target group(s)?</i> Significant outcomes of the project include: 1. More accurate, precise and uniform hardness measurements expressed with realistic and usable measurement uncertainty 2. New generation of hardness measurement instruments (standard, calibration and testing machines) 3. Improved CMCs in the KCDB of the BIPM and enhanced capabilities of the participating NMIs and DIs. 4. Improved CMCs in accreditation scopes of the calibration and testing labs.	<i>What are the expected wider scientific, economic and societal effects of the project contributing to the expected impacts outlined in the work programme and call scope?</i> Scientific: Improved primary level capabilities in hardness measurements. Input into standards related to measurements of hardness reference blocks according to relevant ISO and ASTM Standards. Economic/Technological: Improved measurement capabilities will allow higher throughput leading to cost reduction and lower uncertainties which affects the entire traceability chain. Societal: Increased innovative capacity leading to higher employment and wealth for society. Higher quality and safety of products to be used in daily life

B4 The quality and efficiency of the implementation

This consortium brings together the leading European NMIs, universities and producer companies in high-accuracy hardness metrology field to fix a worldwide problem with *traceability for indentation measurements in Brinell-Vickers-Knoop hardness* project. It is also supported and contributed to by various specialised producer companies in this field that bring in their specific knowledge and experience. The consortium of 15 participants comprises 9 NMIs, 1 research organisation, 2 universities and 3 producer companies, all are leading foundations in their work area in the world of hardness metrology.

B4.a Overview of the consortium

TUBITAK is the Scientific and Technological Research Council of Türkiye. UME (TUBITAK UME) is the NMI of Türkiye operating under the authority of TUBITAK. TUBITAK UME has substantial experience in all metrological activities the field of hardness. It has been accredited to ISO 17025 by TURKAK since 2005, has 14 CMCs in hardness, is participating in CIPM/CCM WGH, ISO TC 164 SC3, IMEKO TC5, EURAMET WG Hardness and accreditation related work in coordination with TURKAK. TUBITAK has experience in designing and development of many primary HSMs and optical measuring systems, currently used as national standards and primary systems. TUBITAK coordinates the project and will contribute to all WPs, and will lead WP4 and WP6.

BEV-PTP is the NMI of Austria, operates a comprehensive quality management system which covers all activities in metrology and fulfils the requirements on the competence of testing and calibration laboratories

of EN ISO/IEC 17025. It cooperates with metrological organisations on the international level in the framework of the Metre Convention, OIML, EURAMET and WELMEC in order to ascertain the worldwide uniformity of measurement units. BEV-PTP is in the process of improving its capabilities in the realisation of hardness, based on recent experiences gained by the design and construction in the field of dead weight FSMs. BEV-PTP will contribute to WP2, WP5 and WP6.

CMI is the NMI of Czechia, and as such ensures the consistency and accuracy of measuring devices and measurements in all fields of scientific, technical and economic activity. CMI ensures the fulfilment of tasks resulting from membership in the EU and in international metrological bodies and organisations (BIPM, CIPM, OIML, EURAMET) and international agreements and arrangements (Metre Convention, CIPM MRA Agreement). CMI's Department of Hardness and Roughness is accredited by the Czech Institute for Accreditation according to the standard EN ISO/IEC 17025:2017 for the calibration of hardness measuring blocks and indentation probes. The Department has three national hardness standards for Rockwell, Vickers and Brinell. CMI will contribute to WP2, WP3, WP5 and WP6.

IMBiH is the NMI of Bosnia and Herzegovina and a self-governing institution under the Council of Ministers in Bosnia and Herzegovina. In accordance with the Law on Metrology of BiH, IMBiH establishes, develops and leads the national metrology system, and represents it in the corresponding international organisations. It is responsible for the enforcement of laws and bylaws and for research and development in the field of metrology. IMBiH was involved in numerous EMRP and EMPIR projects and has CMCs in mass and related quantities. IMBiH will contribute to WP3, WP5 and WP6.

INRIM is a public scientific research body that performs the functions of a NMI for Italy, constituting the presidium of a large part of metrology. INRIM creates, maintains and develops the national reference standards of the seven basic units of the International System (SI) and their respective derived units. Since the 1970s INRIM has been developing its own primary hardness reference standard machines (from micro to macro ranges in all hardness scales for metallic materials) and instrumentation for indenter characterisation which were adopted by several other NMIs in the following years. INRIM has 15 CMCs in hardness, has been the proposer of CIPM/CCM WGH (Chair for 18 years) and is participating in ISO TC 164 SC3, IMEKO TC5, EURAMET TC-M/WG Hardness (coordinator). INRIM has experience in designing and development of many primary HSMs and optical and geometrical measuring systems, currently used as national standards and primary systems in many NMIs and calibration laboratories. INRIM will contribute to all WPs and will lead WP3.

PTB is the NMI of Germany. PTB ensures the traceability of the SI units in legal areas and the fulfilment of tasks resulting from membership in the EU and in international metrological bodies and organisations (BIPM, CIPM, OIML, EURAMET) and international agreements and arrangements (Metre Convention, CIPM MRA Agreement). The Working Group of Hardness Metrology in PTB has substantial experience in all metrological activities in the field of hardness. It is a participant to the international standardisation committee ISO TC 164 SC3, CCM WGH from CIPM (Chair), TC-M from EURAMET and TC 5 from IMEKO. It maintains realisations of all international standardised scales in hardness. For all recommended hardness scales Brinell, Vickers exist CMC entry. PTB has experience optical and tactile surface measuring. PTB will contribute to all WPs and will lead the WP1 and WP2.

GUM is a public scientific research body that performs the functions of a NMI for Poland, constituting the presidium of a large part of metrology. GUM creates, maintains and develops the national reference standards of the seven basic units of the International System (SI) and their respective derived units. It has member status in different metrological organisations: international BIPM (including CCM-WGH), OIML and regional EURAMET, WELMEC, IMEKO (TC5), AIC. GUM maintains a quality management system according to the EN ISO/IEC 17025 and EN 45011. GUM actively cooperates in the field of hardness with other NMIs (i.e. INRIM, CMI) and companies. GUM will contribute to WP2, WP5 and WP6

INM is a public institution in the Republic of Moldova subordinated to the Ministry of Economy and Infrastructure, which implements at national level the policy in the field of metrology by assuring the uniformity, legacy, accuracy and traceability of measurements in the Republic of Moldova. INM conducts its activity based on a Regulation of Organisation and Functioning, focusing on two main directions: legal metrology and applied metrology. INM organises and participates in technical-scientific research of measuring instruments and/or measurements in various areas of public interest. INM is accredited to ISO 17025 by the MOLDAC (Moldavian Accreditation Body) since 2015. Currently, INM is in the process of development of capability. INM will contribute to WP4, WP5 and WP6.

POLITO, the Politecnico di Torino has for 160 years been one of the most prestigious public institutions at both the International and the Italian levels concerning education, research, technological transfer, and services in all sectors of architecture and engineering. The research unit provides the consortium with a strong background in quality engineering and industrial metrology having a long and deep experience on hardness tests. It has focused its current research on the application of artificial intelligence methods for the

geometric and mechanical characterisation of technological surfaces, and the development of simulative approaches for calibration and uncertainty evaluation based on measurement system modelling. POLITO will contribute to WP1, WP2, WP3, WP5 and WP6.

UNZE, University of Zenica, is a public institution with nine organisational units. Among them, Institute "Kemal Kapetanović" deals with scientific and technological research. The Institute has four centres: Centre for metallic materials and technology, Centre for non-metallic materials and technologies, Centre for polytechnic and Centre for environment protection. Five laboratories (three testing and two calibration) are accredited by BATA (Bosnia and Herzegovina body for accreditation). The Mechanical Calibration Laboratory has accreditation since 1998 according to ISO 17025 and has extensive experience in the calibration of HTMs. UNZE will contribute in WP2, WP3, WP5 and WP6.

ZAG is an internationally recognised research organisation that operates in various areas of industry. Several departments deal with sustainable construction, recycling, energy efficiency and cultural heritage, while the Laboratory for Metrology maintains reference standards for force and hardness in Slovenia and performs calibrations. ZAG is accredited to EN ISO/IEC 17025 and ISO 9001 and participates in CIPM/CCM WGH, IMEKO TC5 and in DKD Deutschen Kalibrierdienst. ZAG with significant experience in hardness metrology will undertake leadership of WP5, and will contribute to WP1, WP3 and WP6.

EASYDUR was founded in 1985 with the aim of creating new measuring instruments with the experience already gained in the design and construction of hardness testers and of spring testing. EASYDUR was the first company to fit a camera on hardness testers for Brinell tests. The company has designed and developed state-of-the-art measuring and testing equipment. Today it has a complete range of measuring instruments for tensile-compression, torsion, hardness tests and, last but not least, vision systems for image analysis. The latter have been adopted by national and international customers in a wide range of sectors such as aeronautics, automotive, railways, mills, forges, foundries, and top universities. EASYDUR will contribute to WP3, WP5 and WP6.

LTF has many years of experience in the field of manufacturing, testing and setting-up of measuring instruments, equipment and machine tools for high precision machining. With its reference brand GALILEO® for excellence in Hardness Testing, its Primary Hardness Standardising Machines, developed in close co-operation with the INRIM, have been operating with success for over twenty years in the NMIs of the USA, United Kingdom, Italy, Turkey, Brazil, Bulgaria, China, India, Romania, Russia and Korea. Since 1993, LTF is accredited according to ISO 17025 by ACCREDIA (the Italian Accreditation Body) for calibrations in hardness and length fields. LTF will contribute to WP3, WP5 and WP6.

MPA NRW is the Material Testing Department of North Rhine-Westphalia (Materialprüfungsamt Nordrhein-Westfalen) and has been active in the field of inspection, testing, certification, and calibration since 1947 to support the local economy. Due to many years of activity in the field of calibration of hardness reference blocks, indenters, and hardness testing machines, the MPA has a high level of experience and competence in these areas. The calibration laboratory of the MPA NRW is accredited by the DAkkS (German Accreditation body) according to ISO 17025 for the calibration of hardness reference blocks and indenters, used in hardness testers. The MPA NRW is also participating in the DIN (German standardisation body) and in the ISO TC 164 SC3. MPA NRW will contribute to WP1, WP5 and WP6.

Section C: Detailed project plans by work package

C1 WP1: Development of definition and measurement methodology for hardness indentations of Brinell, Vickers and Knoop

The aim of this work package is to establish precise definitions for the beginning and end points of the indentations made in the application of different hardness testing methods (Brinell, Vickers and Knoop). Due to the testing process, the indentations have no sharp edges. Instead, the indentations gradually merge into the undeformed surface. Sometimes material is pushed out of the indentation, which makes the transition from the undeformed surface to the indentation even more complicated. The currently applied hardness test method determines the size of the indentations using a light microscope. With this method, the curvature of the surface and the size of the optical lens has an influence on whether light reflects back into the lens. In addition, the incidence of light back into the lens also depends on the plane of focus. Furthermore, the influence of sample roughness and illuminance must be examined.

In this work package, there are three tasks:

In Task 1.1, the geometry of possible indentations from the three hardness testing methods, Brinell, Vickers and Knoop, will be investigated and measured to establish a definition for their boundaries.

In Task 1.2, the influence of the selected focal point on the measurement of indentation size will be determined.

In Task 1.3, the influence of the type of lighting and illuminance on the measurement of indentation size will be determined.

C1.a Task 1.1: Definition of indentation boundaries in Brinell, Vickers, and Knoop

The aim of this task is to determine the geometry of the indentations formed by each of the three hardness testing methods so as to arrive at precise definitions of their boundaries. To do this, indentations will be created in materials of differing hardness using the three methods and then measurements of the 3D form of the indentations will be performed. In doing this, the different curvatures of the indents and the points at which the indenter was still in contact with the material during the indentation attempt will be accounted for. Measurements of the indents will be compared with those of the current optical measurement method.

Activity number	Activity description	Participants (Lead in bold)
A1.1.1 M25	PTB, supported by TUBITAK and INRIM will create indents with the three methods (Brinell, Vickers and Knoop) with different forces (for Vickers and Knoop) and different force-diameter indexes (for Brinell) on at least 4 reference blocks of differing hardness. The typical forces for the different methods will be included. The first batch of reference blocks will be produced by M05. Further blocks may be produced up to M25 as necessary for measurement activities.	PTB , TUBITAK, INRIM
A1.1.2 M08	PTB, TUBITAK, INRIM, POLITO, ZAG and MPA NRW will measure the indents on the hardness blocks from A1.1.1 using their own state of the art of hardness measurement systems to compare the results thereby obtained with the current optical method.	PTB , TUBITAK, INRIM, POLITO, ZAG, MPA NRW
A1.1.3 M10	PTB will use scanning tactile and laser confocal measuring systems to measure the 3D surface areas of indentations exhibiting pile-up behaviour that were created with the three methods (Brinell, Vickers and Knoop) on the blocks from A1.1.1. PTB will discuss the results and their implications with TUBITAK, INRIM, POLITO, ZAG and MPA NRW in terms of determining indentation boundaries.	PTB , TUBITAK, INRIM, POLITO, ZAG, MPA NRW
A1.1.4 M10	PTB will use scanning tactile and laser confocal measuring systems to measure the 3D surface areas of indentations exhibiting sink-in behaviour that were created with the three methods (Brinell, Vickers and Knoop) on the blocks from A1.1.1. PTB will discuss results and their implications with TUBITAK, INRIM, POLITO, ZAG and MPA NRW in terms of determining indentation boundaries.	PTB , TUBITAK, INRIM, POLITO, ZAG, MPA NRW
A1.1.5 M15	PTB, supported by TUBITAK, INRIM, POLITO, ZAG and MPA NRW will determine the last point of contact between the indenter and the material from the 3D geometry of the indentations measured in A1.1.3 and A1.1.4.	PTB , TUBITAK, INRIM, POLITO, ZAG, MPA NRW
A1.1.6 M22	PTB, supported by TUBITAK, INRIM, POLITO, ZAG and MPA NRW will analyse the different measurement methods (scanning tactile and confocal laser versus standard	PTB , TUBITAK, INRIM, POLITO,

	optical measuring systems) and results from A1.1.2, A1.1.3 and A1.1.4 in order to generate correction factors for the devices used in industry.	ZAG, MPA NRW
A1.1.7 M25	With input from A1.1.1 – A1.1.6, PTB with TUBITAK, INRIM, POLITO, ZAG and MPA NRW will draft a report on the definition of indentation boundaries in the Brinell, Vickers and Knoop methods. The coordinator, on behalf of PTB, TUBITAK, INRIM, POLITO, ZAG and MPA NRW, will submit the report to EURAMET as deliverable D1 : “Report on the development of universal definitions for Brinell, Vickers and Knoop (BVK) hardness indentation boundaries”.	PTB , TUBITAK, INRIM, POLITO, ZAG, MPA NRW

C1.b Task 1.2: Influence of the focal point on optical indentation size measurements

The aim of this task is to quantify the influence of the focal point on the measured indentation size in order to establish specifications for the focal point in the measurement of indentation sizes. The current lack of specifications of a focal point makes it difficult to compare the indentation size measurements. Therefore, a definition for the correct focus must be established.

Activity number	Activity description	Participants (Lead in bold)
A1.2.1 M15	PTB, supported by TUBITAK, INRIM, POLITO, ZAG and MPA NRW will examine the relationship between varying focal points and measured indentation size in the Brinell method by using their own systems to measure the sizes of at least 10 different indentations on blocks from A1.1.1 in focus and 2-3 measurements below and above the focus point respectively.	PTB , TUBITAK, INRIM, POLITO, ZAG, MPA NRW
A1.2.2 M15	PTB, supported by TUBITAK, INRIM, POLITO, ZAG and MPA NRW will examine the relationship between varying focal points and measured indentation size in the Vickers method by using their own systems to measure the sizes of at least 10 different indentations on blocks from A1.1.1 in focus and 2-3 measurements below respectively above the focus point.	PTB , TUBITAK, INRIM, POLITO, ZAG, MPA NRW
A1.2.3 M15	PTB, supported by TUBITAK, INRIM, POLITO, ZAG and MPA NRW will examine the relationship between varying focal points and measured indentation size in the Knoop method by using their own systems to measure the sizes of at least 10 different indentations on blocks from A1.1.1 in focus and 2-3 measurements below and above the focus point respectively.	PTB , TUBITAK, INRIM, POLITO, ZAG, MPA NRW
A1.2.4 M17	Using measurement results from A1.2.1 – A1.2.3, PTB, with the support of TUBITAK, INRIM, POLITO, ZAG and MPA NRW, will quantitatively establish the relationship between different focal points and indentation size measurements in each of the three methods. Based on this analysis, specifications for most appropriate focal points will be determined for indentation size measurements.	PTB , TUBITAK, INRIM, POLITO, ZAG, MPA NRW

C1.c Task 1.3: Influence of illuminance on optical indentation size measurements

The aim of this task is to investigate the influence of illuminance and related factors on the measured indentation size. The relationship between measured indentation sizes and varying illuminance levels, type of lighting, numerical aperture, and surface roughness will be quantified. The end result will be the establishment of technical specifications for the measurement of indentations sizes in Brinell, Vickers and Knoop measurement systems.

Activity number	Activity description	Participants (Lead in bold)
A1.3.1 M12	PTB, supported by TUBITAK and INRIM will create indents in undeformed surfaces with a minimum of 3 different roughness levels. 10 indents of different diameters will be created on the three selected surfaces with differing roughness for Brinell, Vickers and Knoop. The hardness blocks with the indentations will be circulated among PTB, TUBITAK, INRIM, POLITO, ZAG, and MPA NRW for measurements.	PTB , TUBITAK, INRIM
A1.3.2 M14	PTB, supported by TUBITAK, INRIM, POLITO, ZAG and MPA NRW will determine the difference between indentation size measurements with bright-field versus dark-field illumination by comparing measurements of at least 10 indentations produced in A1.3.1 by each of the three methods under both illumination conditions.	PTB , TUBITAK, INRIM, POLITO, ZAG, MPA NRW

A1.3.3 M14	PTB, supported by TUBITAK, INRIM, POLITO, ZAG and MPA NRW will measure indentations of at least 10 different sizes produced in A1.3.1 in each of the three methods under varying illuminance and examine the results to establish the relationship between illuminance levels and indentation size measurements.	PTB , TUBITAK, INRIM, POLITO, ZAG, MPA NRW
A1.3.4 M16	PTB, supported by TUBITAK, INRIM, POLITO, ZAG and MPA NRW will perform measurements of the indentation sizes on the samples produced in A1.3.1 with varying surface roughness and examine the results to establish the relationship between surface roughness levels and indentation size measurements	PTB , TUBITAK, INRIM, POLITO, ZAG, MPA NRW
A1.3.5 M16	PTB, supported by TUBITAK, INRIM, POLITO, ZAG and MPA NRW will perform measurements on the samples produced in A1.3.1 to examine the influence of the numerical aperture of the lens and optical measurement system on the measured Brinell indentation size in cases of both pile-up and sink-in behaviour.	PTB , TUBITAK, INRIM, POLITO, ZAG, MPA NRW
A1.3.6 M16	PTB, supported by TUBITAK, INRIM, POLITO, ZAG and MPA NRW will perform measurements on the samples produced in A1.3.1 to examine the influence of the numerical aperture of the lens and optical measurement system on the measured Vickers indentation size in cases of both pile-up and sink-in behaviour.	PTB , TUBITAK, INRIM, POLITO, ZAG, MPA NRW
A1.3.7 M18	Using input from A1.2.4, A1.3.2 – A1.3.6, PTB, with the support from TUBITAK, INRIM, POLITO, ZAG and MPA NRW will draft a report document on the development of a measurement methodology to measure indentation sizes in Brinell, Vickers, and Knoop measurement systems taking into account the type of lighting, illuminance, numerical aperture, surface roughness and focal point Once agreed by the consortium, the coordinator, on behalf of TUBITAK, INRIM, POLITO, ZAG and MPA NRW will submit the report to EURAMET as deliverable D2 , <i>“Report on required measurement parameters to be used with the highest accuracy and most commonly used instruments at the NMI level (scanning and optical) for the detection and measurement of indentation dimensions in line with relevant ISO standards”</i>	PTB , TUBITAK, INRIM, POLITO, ZAG, MPA NRW

C2 WP2: Development of transfer standards for Brinell, Vickers and Knoop indentations

Calibration of Indentation Measurement Systems (IMSs) used for Brinell, Vickers and Knoop hardness are performed along the x- and y- axes only using the standard scale and stage micrometer, etc. However, in fact, the geometrical shapes of Brinell, Vickers and Knoop hardness indentations have 3-dimensional (x, y, and z) elements. This variance between the reference used for calibration of IMS and the object to be measured result in systematic errors. A 3-dimensional indentation transfer standard is needed to enable more accurate and reliable calibrations.

The aim of this work package is to develop new transfer standards to be used as a substitute for or concurrent with the current standard scale and stage micrometer for calibrating manual, automatic, semi-automatic IMS of Brinell, Vickers and Knoop hardness scales.

This work package has four tasks:

In Task 2.1, different candidate materials will be investigated for use in indentation transfer standards with varying hardness scales and levels.

In Task 2.2, different designs and materials with varying surface roughness will be investigated for use in indentation transfer standards.

In Task 2.3, the hardness scales and levels to be covered by the indentation transfer standards will be selected.

In Task 2.4 initial samples of the indentation transfer standards will be produced and the repeatability and reproducibility of their measurements will be evaluated.

C2.a Task 2.1: Selection of materials to be used for indentation transfer standards

The aim of this task is to review and select materials to be used for the production of indentation transfer standards for Brinell, Vickers and Knoop hardness. Carbon steel, brass, aluminium, and ceramic etc. will be reviewed as candidate materials for minimum and maximum nominal hardness.

Activity number	Activity description	Participants (Lead in bold)
A2.1.1 M06	PTB, supported by TUBITAK, CMI, INRIM, UNZE and GUM will conduct a literature review on several types of materials including carbon steel, brass, aluminium, and ceramic that may be used for transfer standards for nominal Brinell hardness (100-650) HBW.	PTB , INRIM, TUBITAK, GUM, CMI, UNZE
A2.1.2 M06	PTB, supported by TUBITAK, CMI, INRIM, UNZE and GUM will conduct a literature review on several types of materials including carbon steel, brass, aluminium, and ceramic that may be used for transfer standards for nominal Vickers hardness (100-1000) HV or agreed hardness.	PTB , INRIM, TUBITAK, GUM, CMI, UNZE
A2.1.3 M08	PTB, supported by TUBITAK, CMI, INRIM, UNZE and GUM will conduct a literature review on several types of materials including carbon steel, brass, aluminium, and ceramic that may be used as transfer standards for nominal Knoop hardness (100-1000) HK or agreed hardness.	PTB , INRIM, TUBITAK, GUM, CMI, UNZE
A2.1.4 M10	Using input from A2.1.1 – A2.1.3, INRIM, TUBITAK and PTB will select the material(s) to be used for the production of transfer standards from among the candidate materials studied based on long term stability, desired surface properties and varying hardness levels.	PTB , INRIM, TUBITAK

C2.b Task 2.2: Optimisation of the design of the indentation transfer standards

The aim of this task is to establish the optimum design features of the indentation transfer standards such as optimum shapes (e.g., circle, square, triangle) and thickness, and minimum surface roughness.

Activity number	Activity description	Participants (Lead in bold)
A2.2.1 M10	PTB, supported by TUBITAK, BEV-PTP, CMI, INRIM, GUM and UNZE will perform a review of literature and available data to determine optimum • maximum surface roughness of mechanical surface processing, • block shape (circle, square, triangle), • minimum block thickness, for indentation transfer standards of Brinell hardness.	PTB , INRIM, TUBITAK, GUM, BEV-PTP, CMI, UNZE
A2.2.2 M10	PTB, supported by TUBITAK, BEV-PTP, CMI, INRIM, GUM and UNZE will perform a review of literature and available data to determine optimum • maximum surface roughness of mechanical surface processing, • block shape (circle, square, triangle), • minimum block thickness, for indentation transfer standards of Vickers hardness.	PTB , INRIM, TUBITAK, GUM, BEV-PTP, CMI, UNZE
A2.2.3 M10	PTB, supported by TUBITAK, BEV-PTP, CMI, INRIM, GUM and UNZE will perform a review of literature and available data to determine optimum • maximum surface roughness of mechanical surface processing • block shape (circle, square, triangle), • minimum block thickness for indentation transfer standards of Knoop hardness.	PTB , INRIM, TUBITAK, GUM, BEV-PTP, CMI, UNZE
A2.2.4 M12	With input from A2.2.1 – A2.2.3, INRIM, with the support of TUBITAK and PTB, will produce the design and drawings of the indentation transfer standards together with their mechanical properties for Brinell, Vickers and Knoop hardness scales.	INRIM , TUBITAK, PTB

C2.c Task 2.3: Determination of hardness scales and levels for indentation transfer standards

The aim of this task is to determine optimum hardness scales and levels for the development of indentation transfer standards necessary to cover all possible indentation sizes and shapes that influence measurement results due to their 3-dimensional properties and effects on optical measurement devices. The diameters and diagonal lengths of the indentations on the transfer standards as well as the number and interval of indentations for each sample needed for calibration of hardness indentation measurement systems will also be determined within this task and an overall design for the transfer standards will be produced.

Activity number	Activity description	Participants (Lead in bold)
A2.3.1 M08	PTB, supported by TUBITAK, BEV-PTP, CMI, INRIM, GUM and UNZE will perform an investigation to determine the optimum hardness scale, force-diameter index and hardness level (diameter of indentation) for the transfer standard of Brinell hardness.	PTB , INRIM, TUBITAK, GUM, BEV-PTP, CMI, UNZE
A2.3.2 M10	PTB, supported by TUBITAK, BEV-PTP, CMI, INRIM, GUM and UNZE will perform an investigation to determine the optimum hardness scale and hardness level (diagonal length of indentation) for the transfer standard of Vickers hardness.	PTB , INRIM, TUBITAK, GUM, BEV-PTP, CMI, UNZE
A2.3.3 M10	PTB, supported by TUBITAK, BEV-PTP, CMI, INRIM, GUM and UNZE will perform an investigation to determine the optimum hardness scale and hardness level (diagonal length of indentation) for the transfer standard of Knoop hardness.	PTB , INRIM, TUBITAK, GUM, BEV-PTP, CMI, UNZE
A2.3.4 M12	With input from A2.3.1-A2.3.3, PTB, with support from INRIM and TUBITAK, will determine the hardness scales and levels that are to be covered by the indentation transfer standards for Brinell, Vickers and Knoop indentations so as to optimally meet requirements for the calibration of IMS. A report will be drafted with the conclusions reached. The report will be used as input to A3.4.1.	PTB , INRIM, TUBITAK

C2.d Task 2.4: Production of the indentation transfer standards

The task aims to produce reference indentation blocks to serve as transfer standards for calibrating IMS of Brinell, Vickers and Knoop hardness based on the material and design selections made in Task 2.1 through Task 2.3. The transfer standards produced will be evaluated for repeatability and reproducibility of the results of their measurements.

Activity number	Activity description	Participants (Lead in bold)
A2.4.1 M13	With input from A2.1.4, A2.2.4 and A2.3.4, PTB will procure blocks of the selected materials manufactured according to specifications to produce candidate indentation transfer standards for Brinell, Vickers, and Knoop hardness scales. TUBITAK will realise all selected hardness scales and levels on the different material blocks. The samples will be distributed to PTB, INRIM, TUBITAK, GUM, CMI and UNZE for measurements.	PTB
A2.4.2 M14	PTB, supported by TUBITAK, BEV-PTP, CMI, INRIM, UNZE, and GUM will each measure samples of Brinell hardness indentation transfer standards from A2.4.1 and analyse the results. Based on the results, PTB, TUBITAK CMI, INRIM, UNZE, and GUM will determine the reference values (nominal diameter, diagonal length, measurement uncertainty, etc.) of the indentation transfer standards, validate their available IMSs, and analyse the results.	PTB , INRIM, TUBITAK, GUM, BEV-PTP, CMI, UNZE
A2.4.3 M14	PTB, supported by TUBITAK, BEV-PTP, CMI, INRIM, UNZE, and GUM each measure samples of Vickers hardness indentation transfer standards from A2.4.1 and analyse the results. Based on the results, PTB, TUBITAK, CMI, INRIM, UNZE, and GUM will determine the reference values (nominal diameter, diagonal length, measurement uncertainty, etc.) of the indentation transfer standards, validate their available IMSs and analyse the results.	PTB , INRIM, TUBITAK, GUM, BEV-PTP, CMI, UNZE
A2.4.4 M14	PTB, supported by TUBITAK, BEV-PTP, CMI, INRIM, UNZE, and GUM each measure reference samples of Knoop hardness from A2.4.1 and analyse the results. Based on the results, PTB, TUBITAK, CMI, INRIM, UNZE, and GUM determine the reference values (nominal diameter, diagonal length, and measurement uncertainty, etc.) of the indentation transfer standards, validate their available IMSs, and analyse the results.	PTB , INRIM, TUBITAK, GUM, BEV-PTP, CMI, UNZE
A2.4.5 M15	PTB supported by INRIM, TUBITAK, BEV-PTP and GUM will analyse results from A2.4.2-A2.4.4 to select a subset of the candidate transfer standards based on criteria for needed diameters, minimum uncertainty etc. The subset will be circulated to PTB, INRIM, TUBITAK, BEV-PTP and GUM for additional measurements.	PTB , INRIM, TUBITAK, GUM, BEV-PTP

A2.4.6 M16	PTB, TUBITAK, BEV-PTP, INRIM and GUM will each measure the prototype transfer standards of Brinell, Vickers and Knoop hardness selected in A2.4.5 and analyse the results to determine the repeatability and reproducibility of measurements performed on the standards.	PTB , INRIM, TUBITAK, GUM, BEV-PTP
A2.4.7 M18	Using input from A2.4.1 – A2.4.6 PTB, with the support of INRIM, TUBITAK, BEV-PTP, GUM, CMI, and UNZE, PTB will draft a report on the development and performance evaluation of reference Brinell, Vickers and Knoop indentation standards. Once agreed, the coordinator, on behalf of PTB, INRIM, TUBITAK, BEV-PTP, GUM, CMI, and UNZE will submit the report to EURAMET as D3 “ <i>Report on the development and validation of reference Brinell, Vickers and Knoop indentations to be used as transfer standards</i> ”	PTB , INRIM, TUBITAK, GUM, BEV-PTP, CMI, UNZE

C3 WP3: Traceability chain from NMI to user level

The aim of this work package is to provide a detailed methodology to ensure a robust traceability chain for Brinell, Vickers and Knoop hardness tests from NMI to end users in testing facilities. The currently available traceability chain covers only uniaxial static forces, applied perfectly through a known indenter, at constant temperature, to test specimens that react in a purely axisymmetric manner. This is not how systems perform in practice, where there are additional effects due to short-term creep resulting from different force application times, indenter geometry, temperature, alignment, optical measurements on the indentation size and fabrication of the artefacts used as transfer standards. For this reason, it is necessary to investigate all possible applications of Brinell, Vickers and Knoop hardness tests at user and industrial levels and the main instrumentation used to perform such tests and to calibrate them according to international standards.

In Task 3.1, a comprehensive description of the existing hardness calibration and testing infrastructure, covering both physical and documentary standards will be developed.

In Task 3.2, the effects of the main influencing factors in Brinell, Vickers and Knoop hardness during tests and calibrations at all levels of the traceability chain will be identified and evaluated.

In Task 3.3, an advanced model with associated uncertainty estimations will be developed.

In Task 3.4, a robust traceability chain for Brinell, Vickers and Knoop hardness measurements from NMI to user level for metrological services and industrial tests will be established.

C3.a Task 3.1: Review of hardness testing machines, standards, and applications

The aim of this task is to review the technology and procedures currently used for metrological services in the field of Brinell, Vickers and Knoop hardness tests at all levels from NMI to end users and to identify industrial hardness measurement applications not currently addressed by existing calibration technology.

Activity number	Activity description	Participants (Lead in bold)
A3.1.1 M05	INRIM, TUBITAK, CMI, IMBIH, PTB, POLITO, UNZE, ZAG, EASYDUR and LTF will perform a review of commercially available Brinell, Vickers and Knoop hardness testing and calibration machines at industrial levels. Information on the measurement principles, force range, stiffness, indenter geometry, dynamic characteristics, optical bench will be collected. Associated instrumentation will also be considered during the review. In total, 5 different types of commercially available hardness testing machines for each scale will be reviewed with each participant providing details of at least one.	INRIM , TUBITAK, CMI, IMBIH, PTB, POLITO, UNZE, ZAG, EASYDUR, LTF
A3.1.2 M06	INRIM, TUBITAK, CMI, IMBIH, PTB, ZAG, MPA NRW and LTF will perform a review of commercially available Brinell, Vickers and Knoop calibration machines at NMI and calibration laboratory levels. Information on the measurement principles, force range, stiffness, indenter geometry, dynamic characteristics, optical bench will be collected. Associated instrumentation will also be considered during the review. At least, 5 different types of calibration machines for each scale will be reviewed.	INRIM , TUBITAK, CMI, IMBIH, PTB, ZAG, MPA NRW, LTF
A3.1.3 M07	INRIM, TUBITAK, CMI, IMBIH, PTB, POLITO, UNZE, ZAG, EASYDUR and LTF will each perform a review of national and international standards, recommendations, and procedures for Brinell, Vickers and Knoop hardness tests and calibration of machines in their own country to understand what is available. Each participant will produce a list of the national and international standards adopted in their own country.	INRIM , TUBITAK, CMI, IMBIH, PTB, POLITO, UNZE, ZAG, EASYDUR, LTF

A3.1.4 M08	Based on the list of standards obtained from A3.1.3, INRIM, TUBITAK, PTB, BEV-PTP, CMI, IMBIH, UNZE, ZAG and LTF will examine in detail international standards and documents such as ISO 6506, ISO 6507, ISO 4545 series, ASTM E92-17, ASTM E10-18, etc. and, if they exist, their own national standards and documents.	INRIM, TUBITAK, PTB, BEV-PTP, CMI, IMBIH, UNZE, ZAG, LTF
A3.1.5 M10	INRIM, TUBITAK, CMI, IMBIH, PTB, POLITO, UNZE, ZAG, EASYDUR and LTF will select representative industrial applications, such as automotive, offshore, pipeline, oil and gas, structural, manufacturing, aerospace, agriculture, mining, pulp and paper, nuclear, power generation, and more, where Brinell, Vickers and Knoop hardness tests (taking into account traceability) are performed as the most widely used test for evaluating the mechanical properties of metals and other materials. The industrial applications will be selected in order to cover a wide variety of materials and scales, chosen from the about 25 Brinell, 20 Vickers and 12 Knoop standardised scales covering, each, different levels of hardness: from very soft to very hard materials (aluminium, copper, brass, cast iron, steel, carbon steel, etc.).	INRIM, TUBITAK, CMI, IMBIH, PTB, POLITO, UNZE, ZAG, EASYDUR, LTF
A3.1.6 M12	Using input from A3.1.1 to A3.1.5, INRIM with support from TUBITAK, CMI, IMBIH, PTB, POLITO, UNZE, ZAG, EASYDUR and LTF will produce a report on the review of: - commercially available Brinell, Vickers and Knoop hardness testing and calibration machines, - implemented national and international standards, - industrial applications where Brinell, Vickers and Knoop hardness tests are performed. The report will be used in order to focus the activities of Task 3.2 on the most used hardness scales, materials and applications.	INRIM, TUBITAK, CMI, IMBIH, PTB, POLITO, UNZE, ZAG, EASYDUR, LTF

C3.b Task 3.2: Investigation of influencing factors in Brinell, Vickers and Knoop hardness tests

The aim of this task is to investigate a range of factors influencing BVK hardness tests, such as:

- test piece/material: test piece thickness, stiffness of the support, coarseness of grain structure, surface roughness, inhomogeneous distribution of hardness, surface cleanliness, etc.
- testing machine: relating to the a) machine frame (friction loss, elastic deflection, misalignment of the indenter holder, etc.) b) measuring system (indicating error, poor resolution, numerical aperture of lens or illuminator, inhomogeneous illumination of the indentation, etc.) c) force application system (deviation from nominal forces, deviation from time intervals of the testing cycle, test force dwell time, force introduction, overrun of test forces, indentation velocity, etc.) d) indenters (deviation from the ideal shape, indenter holder, damage, deformation under force, etc.)
- environment: deviation and/or drift of ambient temperature, vibration and shocks, etc.
- operator: wrong selection of test method, handling, reading, evaluation errors, etc

Some of these factors have already been studied and their effect can be derived from data that can be found in the literature: therefore, in this task, the following factors on which such information is only partially known or not known will be studied and investigated:

- short term creep
- indenter's geometry
- temperature
- alignment of applied load.

Hardness tests, while considered non-destructive, cannot be replicated exactly in the same point as they plastically deform the measuring point. It is therefore necessary to use specially prepared hardness blocks in which the homogeneity of the surface hardness is sufficiently guaranteed so as to be able to highlight significant variations in hardness to be attributed to the variation of the test parameters and not to the variation in hardness of the material. These hardness blocks are expensive and have a limited surface area, so it is necessary to optimise the test procedures regarding the hardness scales to be examined, hardness values, number and position of the indentations on the block surfaces, etc.

Participants will divide the experiments on the different scales and levels of hardness according to their measurement capabilities and availability of hardness blocks and indenters, in order to cover the entire experimental plans.

Activity number	Activity description	Participants (Lead in bold)
A3.2.1 M16	INRIM, TUBITAK, CMI and PTB will design the optimal experimental plan and the materials on which the Brinell, Vickers and Knoop hardness tests are to be performed. The choice of the experimental plan will have to take into account possible correlations between the individual factors and will have to include theoretical exercises and investigations, simulations in a standard machine, actual measurements performed in testing machines or combination of these.	INRIM , TUBITAK, CMI, PTB
A3.2.2 M20	INRIM, TUBITAK, CMI, and PTB will investigate the effects of <i>short-term creep</i> on the results of the selected hardness tests according to the experimental plan from A3.2.1. The experimental workload will be divided among INRIM, TUBITAK, CMI and PTB on the basis of their measurement possibilities (different BVK scales) and on the availability of blocks made with different materials. However, being able to count on a strong overlap of these available scales and materials by the participants, INRIM will guarantee an extensive coverage of the adopted experimental plan.	INRIM , TUBITAK, CMI, PTB
A3.2.3 M20	INRIM, TUBITAK, CMI, IMBIH, UNZE, ZAG and PTB will investigate the effects of <i>indenters' geometry</i> on the results of the selected hardness tests according to the experimental plan from A3.2.1. The experimental workload will be divided among INRIM, TUBITAK, CMI, IMBIH, UNZE, ZAG and PTB on the basis of their measurement possibilities (different BVK scales) and on the availability of blocks made with different materials. However, being able to count on a strong overlap of these available scales and materials by the participants, INRIM will guarantee an extensive coverage of the adopted experimental plan.	INRIM , TUBITAK, CMI, PTB, IMBIH, UNZE, ZAG
A3.2.4 M20	INRIM, TUBITAK, CMI and PTB will investigate the effects of <i>temperature</i> on the results of the selected hardness tests according to the experimental plan from A3.2.1. The experimental workload will be divided among INRIM, TUBITAK, CMI and PTB on the basis of their measurement possibilities (different BVK scales) and on the availability of blocks made with different materials. However, being able to count on a strong overlap of these available scales and materials by the participants, INRIM will guarantee an extensive coverage of the adopted experimental plan.	INRIM , TUBITAK, CMI, PTB
A3.2.5 M20	INRIM, TUBITAK, CMI, and PTB will investigate the effects of <i>alignment of the applied load</i> on the results of the selected hardness tests according to the experimental plan from A3.2.1. The experimental workload will be divided among INRIM, TUBITAK, CMI and PTB on the basis of their measurement possibilities (different BVK scales) and on the availability of blocks made with different materials. However, being able to count on a strong overlap of these available scales and materials by the participants, INRIM will guarantee an extensive coverage of the adopted experimental plan.	INRIM , TUBITAK, CMI, PTB
A3.2.6 M24	Using input from A1.2.4, A1.3.7, A3.1.6 and from A3.2.2 - A3.2.5, INRIM, with support from TUBITAK, CMI and PTB will produce a report summarising the potential effect of various influence factors (i.e. focal point, illuminance, short-term creep, indenters' geometry, temperature, alignment of applied load) and their relative magnitudes on the results of a range of Brinell, Vickers and Knoop hardness tests. The report will be used as a contribution to EURAMET TC-M / WG on Hardness for developing an open access database containing the sensitivity coefficients of the different influence parameters in Brinell, Vickers and Knoop hardness measurements, to which those already known on other scales of hardness can be added. This database, which can be constantly updated, will serve the users as a fundamental support for the evaluation of uncertainty in hardness measurements.	INRIM , TUBITAK, CMI, PTB

C3.c Task 3.3: Advanced uncertainty models for Brinell, Vickers and Knoop hardness tests

The aim of this task is to develop uncertainty models describing the effects of the various application conditions in relation to the different types of hardness testing machines at NMI, calibration laboratory and user levels and other associated effects on the measurement results of Brinell, Vickers and Knoop hardness tests. The models will be parameterised with the magnitudes of the sensitivity coefficients known from the

literature and also determined from earlier work in the project (Task 1.2, Task 1.3, Task 3.2), resulting in realistic uncertainty evaluations.

Activity number	Activity description	Participants (Lead in bold)
A3.3.1 M25	Using input from A3.2.6 and with specific reference to ISO 6506, ISO 6507, ISO 4545 series, INRIM, with support from TUBITAK, CMI, PTB, IMBIIH, POLITO, UNZE and ZAG, will determine the main sources of uncertainty associated with the selected hardness tests from among representative scales and hardness values among all 25 Brinell, 20 Vickers and 12 Knoop scales.	INRIM , TUBITAK, CMI, PTB, IMBIIH, POLITO, UNZE, ZAG
A3.3.2 M26	Using input from A3.3.1 and with specific reference to ISO 6506, ISO 6507, ISO 4545 series, INRIM, with support of TUBITAK, CMI, PTB, IMBIIH, POLITO, UNZE and ZAG will develop advanced measurement uncertainty models for hardness tests based on the major uncertainty components determined in A3.3.1.	INRIM , TUBITAK, CMI, PTB, IMBIIH, POLITO, UNZE, ZAG
A3.3.3 M27	Using input from A3.3.2 and with specific reference to ISO 6506, ISO 6507, ISO 4545 series, INRIM, with support of TUBITAK, CMI, PTB, IMBIIH, POLITO, UNZE and ZAG, will verify the measurement uncertainty model developed in A3.3.2 by testing in a representative range of hardness testing machines, both including different hierarchy levels (NMI, calibration laboratories of hardness blocks and hardness machines, industrial testing machines) and different type of scales (Brinell, Vickers and Knoop).	INRIM , TUBITAK, CMI, PTB, IMBIIH, POLITO, UNZE, ZAG
A3.3.4 M30	Using input from A3.3.3, INRIM, with support of TUBITAK, CMI, PTB, IMBIIH, POLITO, UNZE and ZAG will draft a report on the validation of the uncertainty model for Brinell, Vickers and Knoop hardness tests. Once the consortium has agreed on the validation report, the coordinator on behalf of INRIM, TUBITAK, CMI, PTB, IMBIIH, POLITO, UNZE and ZAG, will then submit the report to EURAMET as D4 : “Report on the validation of advanced uncertainty models developed for Brinell, Vickers and Knoop hardness tests”. The report will be used as a contribution for revising the calibration guide EURAMET cg-16, Version 2.0 (03/2011): <i>Guidelines on the Estimation of Uncertainty in Hardness Measurements</i> , which currently covers Rockwell scales only.	INRIM , TUBITAK, CMI, PTB, IMBIIH, POLITO, UNZE, ZAG

C3.d Task 3.4: Development of a traceability chain methodology for Brinell, Vickers and Knoop hardness tests

The aim of this task is to develop a methodology to establish a traceability chain for Brinell, Vickers and Knoop hardness tests from NMI to user level.

In a wide range of industrial applications there is the need to measure hardness. In each application, there will be an uncertainty requirement on the hardness measurement – the equipment used to make the measurement must be traceable to a primary realisation of the hardness scales within this required uncertainty. The situation may vary slightly from country to country, but a correct traceability should be based on a country having one national metrology institute (NMI) realising the hardness scales in a number of primary hardness standard machines used to calibrate primary hardness reference blocks and a number of calibration laboratories, generally accredited by their national accreditation body, using hardness calibration machines to calibrate hardness reference blocks. These blocks may then be used to calibrate industrial hardness testing machines. The hardness calibration machines will generally be traceable to the national hardness standard machines via comparisons using hardness blocks as transfer standards – and the accredited calibration and measurement capability (CMC) of the calibration laboratory will be based on the results of these comparisons. Calibration of hardness blocks in the hardness calibration machines will generally be carried out in accordance with a documented procedure, such as part 3 of the related ISO standards, and the uncertainty of the calibration results will be dependent on the machine’s CMC, as well as on the performance of the instrument during the calibration. Similarly, the uncertainty of the calibration of the industrial hardness testing machines will be partly dependent on the uncertainty arising from the hardness reference blocks used for their calibration, and the uncertainty of any subsequent hardness measurement will depend in part on the uncertainty associated with the hardness testing machine. It can be seen that the uncertainty of the final hardness measurement is dependent on all of the previous traceability stages, and a clear and harmonised guidance on how to estimate all of these contributions is needed since at present it is partially, and sometimes with different approaches, given in annexes to some related national and international standards (i.e. ISO or ASTM standards) and in some international documents (i.e. EURAMET cg-16). The above traceability situation strictly covers only indirect calibration and verification, whereas direct traceability to the SI quantities involved in the hardness tests (force, length, time) should also be addressed.

Activity number	Activity description	Participants (Lead in bold)
A3.4.1 M32	Using input A1.1.7, A1.3.7, A2.3.4, A3.1.6, A3.2.6, and A3.3.4, INRIM, with support of TUBITAK, CMI, PTB, IMBIH, POLITO, UNZE and ZAG, will develop improved calibration and test procedures extending the present traceability chain for Brinell, Vickers and Knoop hardness tests from NMI to user level, ensuring that all influencing factors identified in A3.2.5 are properly considered by the calibration and test procedures.	INRIM , TUBITAK, CMI, PTB, IMBIH, POLITO, UNZE, ZAG
A3.4.2 M35	With input from A3.4.1, INRIM, with the support of TUBITAK, CMI, PTB, IMBIH, POLITO, UNZE and ZAG will draft guidelines for the establishment of traceability for Brinell, Vickers and Knoop hardness test measurements. The scope of the guideline will be to give guidance on the steps necessary to assure traceability of hardness measurements and on the estimation of hardness measurement uncertainty for each of these steps, namely: • uncertainty of hardness measurements carried out by national hardness standard machines and uncertainty of the primary hardness reference blocks calibration • uncertainty of hardness measurements carried out by hardness calibration machines (i.e. determination of CMCs) and uncertainty of the hardness reference blocks calibration • uncertainty of hardness measurements carried out by industrial hardness testing machines • In each of these cases, the uncertainty determination is based on two major components – the uncertainty obtained during the direct and indirect calibration of the equipment and the uncertainty resulting from the equipment's subsequent use. Once agreed by the consortium, the coordinator on behalf of INRIM, TUBITAK, CMI, PTB, IMBIH, POLITO, UNZE, and ZAG, will then submit the guidelines to EURAMET as D5 , “ <i>Guidelines for the establishment of traceability in Brinell, Vickers and Knoop hardness test measurements from NMI to end user level</i> ”.	INRIM , TUBITAK, CMI, PTB, IMBIH, POLITO, UNZE, ZAG
A3.4.3 M35	Based on input from A3.4.2, INRIM, TUBITAK, PTB, EASYDUR, and LTF will draft a set of recommendations for hardness instrument producers aimed at facilitating the development of the next generation of hardness testing machines for Brinell, Vickers and Knoop hardness measurements. Such recommendations will serve the instrument producers to update hardness machines to comply with updated documentary standards and the requirements for improved calibration and testing procedures. Recommendations will include technical specifications for the instrumentation so that they may be properly calibrated for assurance of traceable measurements. Once agreed by the consortium, the coordinator on behalf INRIM, TUBITAK, PTB, EASYDUR, and LTF will submit the recommendations to EURAMET as D6 , “ <i>List of recommendations for hardness instrument producers including technical specifications</i> ”.	INRIM , TUBITAK, PTB, EASYDUR, LTF

C4 WP4: Automatisisation of indentation measurements

National Metrology Institutes (NMIs) with Brinell, Vickers and Knoop calibration and measurement capabilities (CMCs) perform the indentation measurements in different ways; some already use custom made software to perform automatic and semi-automatic measurements, others realise measurements manually with the operator deciding the location of border of indentation by visual inspection. In the current automatic measuring systems at NMIs, there is a single measurement axis, and the border of indentation is determined by the software and an automatic measurement is taken in one direction only. The perpendicular length (diameter/diagonal length) measurement is made by rotating the indentation around its axis and the second diameter and/or diagonal length is measured in the same way. This might therefore be called a kind of semi-automatic measurement, and this is possible only for one indentation measurement; there is no possibility to perform measurements in series on more than one indentation automatically.

This work package will investigate the possibility of performing automatic measurements of the diameter of circular shaped Brinell indentations, the diagonal length of square shaped Vickers indentations and the diagonal length of equilateral shaped Knoop indentations. This will be done by making the necessary mechanical modifications to and developing fit for purpose automation software for use of IMS present in TUBITAK's Hardness Laboratory. A apparatus for marking the locations of the indentation, where the

indentations and their measurements are to be realised, and an algorithm to enable automatic indentation measurement in both directions, namely X and Y, will be developed and integrated into the system. This will be implemented within an IMS with an existing four-axis (X, Y, Z and R (rotation)) motorised stage. The main features of the targeted improved system include:

- auto-focusing ability,
- recording the coordinates of the indentation locations placed by the operator over the surface of the hardness reference block / certified reference material (CRM)
- marking the locations on the surface with various diameters depending on the size of indentations
- marking the positioning of the block on to the IMS to replace it after realisation of indentations
- measurement of indentations after their realisation

Not all indentations are suitable for automatic measurement due to optical restrictions. The effective parameters are expected to be the lack of contrast, optical resolution, measurement conditions (lighting etc), shape and quality of indentation etc. At the end of this work package different types of indentations on various material will be analysed and the indentations suitable for automatic measurements will be determined and will be presented as a result of this project. Following from this, automatic indentation measurement systems will be used for at least this type of indentations.

This work package consists of four tasks:

In Task 4.1, a review will be performed on current indentation measurement systems and their software.

In Task 4.2, mechanical modifications will be made to the IMS at TUBITAK to add the capability to determine and mark the location of indentations.

In Task 4.3, software will be developed for automatic indentation measurements.

In Task 4.4, analysis of indentations will be done and characteristics of indentations suitable for automatic measurements will be determined.

C4.a Task 4.1: Review the current indentation measurement systems and software

The aim of this task is to undertake a review of the present IMSs at the highest accuracy level and the indentation measurement units of hardness machines, their working principles, electro-mechanical structure, and integrated software, if any. Information obtained from this review will be used as input for designing and developing the necessary modifications and software to improve TUBITAK's system.

Activity number	Activity description	Participants (Lead in bold)
A4.1.1 M06	TUBITAK, BEV-PTP, INRIM, PTB and INM will review the automatic indentation measurement systems (either primary reference systems, or calibration or testing machines) and relevant software, including measurement algorithms, currently used in these systems for Brinell indentation measurement in industrialised countries.	TUBITAK , BEV-PTP, INRIM, PTB, INM
A4.1.2 M08	TUBITAK, BEV-PTP, INRIM, PTB and INM will review the automatic indentation measurement systems (either primary reference systems or testing machines) and relevant software, including measurement algorithms, currently used in these systems for Vickers indentation measurement in industrialised countries.	TUBITAK , BEV-PTP, INRIM, PTB, INM
A4.1.3 M10	TUBITAK, BEV-PTP, INRIM, PTB and INM will review the automatic indentation measurement systems (either primary reference systems or testing machines) and relevant software, including measurement algorithms, currently used in these systems for Knoop indentation measurement in industrialised countries.	TUBITAK , BEV-PTP, INRIM, PTB, INM
A4.1.4 M12	TUBITAK, BEV-PTP, INRIM, PTB and INM will review the automatic indentation measurement systems (either primary reference systems or testing machines) and relevant software used in these systems for auto-calibration of the system by making use of a certified line scale present in use in industrialised countries.	TUBITAK , BEV-PTP, INRIM, PTB, INM
A4.1.5 M12	Using input from A4.1.1 – A4.1.4, TUBITAK, BEV-PTP, INRIM, PTB and INM will prepare a short report in which the results from the review of the IMSs will be summarised. The main parameters for realisation of measurements made by the current systems will be determined and their performances will be evaluated in terms of accuracy and user-friendliness.	TUBITAK , BEV-PTP, INRIM, PTB, INM

C4.b Task 4.2: Modifying the present system at TUBITAK to enable marking of indentation locations

The aim of this task is to implement improvements to the present IMS at TUBITAK's Hardness Laboratory in order to endow it with the capability for marking the locations of indentations that are selected by the operator from the software of the IMS. This is foreseen to be made automatically by the IMS itself after the operator defines the location of the indentations that are to be realised.

Activity number	Activity description	Participants (Lead in bold)
A4.2.1 M08	TUBITAK with the help of INRIM and PTB will investigate the possibility of mounting a marking tool onto the IMS currently available at TUBITAK's Hardness Laboratory. In the case that such a system/tool is not available for adaptation, a new one will be designed. The tool would be used for marking the hardness reference blocks with the specified indentation diameters in accordance with the hardness scale and level, the targeted coordinates and number of indentations. Additionally, the modified IMS should have the capability to mark the correct position of the hardness block on the IMS plate for indentation measurements after realisation of indentations (to ensure identical positioning of the block for marking the location of indentations and for measurement after realisation of indentations).	TUBITAK , INRIM, PTB
A4.2.2 M11	Using input from A4.2.1, TUBITAK will buy or produce the marking tool and the apparatus will be mounted to the IMS.	TUBITAK
A4.2.3 M17	TUBITAK with the help of INRIM and PTB will test the functionality of the modified IMS by performing automatic marking of hardness blocks with specified indentation diameters at the targeted locations by integrating the mounted mechanical parts with the software of the system.	TUBITAK , INRIM, PTB

C4.c Task 4.3: Improvement of the TUBITAK system software

The aim of this task is to make improvements to the software of TUBITAK's IMS to enable the following for indentations suitable for automatised measurements:

- auto focusing
- recording indentation locations in system memory
- marking the locations on the surface with specified indentation diameters
- marking the correct position of the hardness block on IMS plate for indentation measurements after realisation of indentations
- measurement of indentations after their realisation

Activity number	Activity description	Participants (Lead in bold)
A4.3.1 M21	With input from A4.1.5, TUBITAK, INRIM, and PTB will determine specifications for the improvements to the software of the IMS currently available at TUBITAK Hardness Laboratory to enable autofocus, edge detection by image processing for circular shaped indentations, automatic measurement of one indentation and serial automatic measurements to n number of Brinell indentations whose coordinates were already recorded by the software during marking for indentations suitable for automatised measurements.	TUBITAK , INRIM, PTB
A4.3.2 M24	With input from A4.1.5, TUBITAK, INRIM, and PTB will determine specifications for the improvements to the software of the IMS currently available at TUBITAK's Hardness Laboratory to enable autofocus, edge detection by image processing for square shaped indentations, automatic measurement of one indentation and serial automatic measurements to n number of Vickers indentations whose coordinates were already recorded by the software during marking for indentations suitable for automatised measurements.	TUBITAK , INRIM, PTB

A4.3.3 M27	With input from A4.1.5, TUBITAK, INRIM, and PTB will determine specifications for the improvements to the software of the IMS currently available at TUBITAK's Hardness Laboratory to be able to make autofocus, edge detection by image processing for equilateral shaped indentations, automatic measurement of one indentation and serial automatic measurements to n number of Knoop indentations whose coordinates were already recorded by the software during marking for indentations suitable for automatised measurements.	TUBITAK, INRIM, PTB
A4.3.4 M30	TUBITAK will subcontract the work of developing the improved software to be implemented in the IMSs according to the software improvement specifications determined in A4.3.1 – A4.3.3. Subcontracting is necessary as the consortium does not have the necessary skills / expertise for this work. The cost is expected to be €12 500.	TUBITAK
A4.3.5 M30	TUBITAK, with the support of INRIM, and PTB will test the software developed in A4.3.4 by comparing the measurement results within the same system between automatic and manual measurements as well as between different automatic IMSs. Feedback will be given to A4.3.4 if further improvements are necessary in the software.	TUBITAK, INRIM, PTB
A4.3.6 M32	With input from A4.3.4 and A4.3.5, TUBITAK, INRIM, and PTB will draft a user manual for the developed software. Once agreed by the consortium, the coordinator, on behalf of TUBITAK, INRIM and PTB, will submit the user manual to EURAMET as D7 , “ <i>User manual for the software/algorithm developed for automated BVK indentation measurements</i> ”.	TUBITAK, INRIM, PTB

C4.d Task 4.4: Analysis of indentations suitable for automatic measurements

The aim of this task is to perform research on the types of indentations suitable for automatic measurements and define the materials, indentation shapes, hardness levels and scales, and optical systems required for the performance of automated measurements.

Activity number	Activity description	Participants (Lead in bold)
A4.4.1 M30	With input from A4.1.5, A4.3.4-A4.3.5, TUBITAK, INRIM and PTB will study the optical principles that an indentation measurement system is working with. The optical specifications of the material (hardness reference blocks), the optical devices, and the theoretical approach will be investigated to determine how the optical system together with the software recognises the surface during autofocus and the indentation border with the aim of identifying optical limits for recognition.	TUBITAK, INRIM, PTB
A4.4.2 M33	TUBITAK, INRIM and PTB will study different indentation shapes in relation to hardness levels and scales and the material of the hardness reference blocks. In addition, the indents realised in A2.4.1, if needed, more differently sized of indentations will be realised on different types of blocks by making use of the hardness standardising machines available at the TUBITAK's Hardness Laboratory.	TUBITAK, INRIM, PTB
A4.4.3 M35	INRIM and PTB will make measurements of the indentations realised in A2.4.1 and A4.4.2 on their own systems using different types of lenses and adjustments. These results will be compared with automatised measurements of the same indentations made at TUBITAK and a correlation between the indentation specifications and accuracy of automatic measurements will be determined. The purpose is to identify the characteristics of indentations that are suitable for automatised measurements.	TUBITAK, INRIM, PTB
A4.4.4 M36	With input from A4.4.1 – A4.4.3 TUBITAK, INRIM and PTB will draft a report on the analysis of the results of automatic Brinell, Vickers and Knoop indentation measurements with specifications for the types of indentations suitable for automatic measurements. Once agreed by the consortium, the coordinator, on behalf of TUBITAK, INRIM and PTB will submit the report to EURAMET as D8 “ <i>Report on the analysis of the results of automatic Brinell, Vickers and Knoop indentation measurements with specifications for the types of indentations suitable for automatic measurements</i> ”.	TUBITAK, INRIM, PTB

C5 WP5: Creating impact**C5.a Task 5.1: Dissemination and communication**

Activity number	Activity description	Participants (Lead in bold)
A5.1.1 M36	At the beginning of the project the consortium will constitute a Stakeholder Committee (SC) with at least 5 members, at least 3 of which will be from European countries, including three groups of stakeholders: the national standardisation technical committees, accredited calibration laboratories and manufacturers of hardness instruments (standardising, calibration and testing machines). The aim of the SC is to clarify stakeholder needs to feed them into the project and also to disseminate project results to stakeholders in order to create an impact. Interaction with the SC will be achieved mostly by teleconferences regularly held in conjunction with the project meetings and emails. A meeting in person will be held in conjunction with the M18 project meeting. Additional interaction and communication with stakeholders will be through individual meetings between SC members and the project participants and through activities such as participation in technical committee meetings (e.g. EURAMET TC-M Working Group on Hardness, IMEKO TC5, CCM Working Group on Hardness), conferences and workshops. Furthermore, communication between the project participants and the SC will be carried out via email and through the project website. TUBITAK will organise the teleconferences and keep the register of the SC and send periodic short communications on the project outcomes.	TUBITAK , all participants
A5.1.2 M36	A project website will be created and hosted by TUBITAK with public access in general and a restricted section for participants only. The website will be managed and regularly updated (at minimum every 6 months) by TUBITAK with information such as project reports, papers published by the participants, project meetings and project events such as trainings and workshops. Project outputs such as public reports, presentations, etc. will be made publicly available. The restricted section of website will be accessible to project participants only for the exchange of data, documents, and information. Each project participant will provide publishable information about their project involvement to TUBITAK to be uploaded on the website. The project website will remain live for as long as possible (at least 3 years after the end of the project). The project website will clearly acknowledge, in a prominent position on the homepage (e.g. in the header, footer or centre and in a readable size), the Metrology Partnership. This will be done by including either i) the Partnership project website header/footer badge and acknowledgement text (this text can be anywhere on homepage) or ii) the Partnership acknowledgement badge.	TUBITAK , all participants
A5.1.3 M36	The participants will submit 4 papers to peer-reviewed journals during the project. At least 3 of the 4 open access publications will be the result of a collaborative effort from participants from different countries. Target journals include but not limited to: • Measurement Science and Technology • Measurement • Acta IMEKO • Measurement: sensor Another route for publication is also offered by the "Open Research Europe" initiative launched by the European Commission. The authors of the open access peer-reviewed papers will clearly acknowledge the financial support provided through the Partnership as required by EURAMET in accordance with Article 17, Article 18 and Annex 5 of the Grant Agreement with the following text: “The project 22RPT01 TracInd BVK-H has received funding from the European Partnership on Metrology, co-financed from the European Union’s Horizon Europe Research and Innovation Programme and by the Participating States.” The authors will ensure that the following meta data is submitted and included for each paper: • Funder name: European Partnership on Metrology • Funder ID: 10.13039/100019599	ZAG , all participants

	Grant number: 22RPT01 TracInd BVK-H																			
A5.1.4 M36	The participants will present at least 4 oral or poster presentations at the following international conferences: - XXIV IMEKO World Congress, Hamburg, 2024 - IMEKO TC5 Conference (not yet determined, tentatively planned for after June 2025) Further relevant conferences may be identified during the project	ZAG , all participants																		
A5.1.5 M36	Information on the results of the project will be disseminated to a range of standards bodies and committees and feedback will be sought (see details below). <table><tr><th>Standards Committee / Technical Committee / Working Group</th><th>Participants involved</th><th>Likely area of impact / activities undertaken by participants related to standard / committee</th></tr><tr><td>ISO/TC164 SC 3 and WG 2</td><td>PTB, INRIM, TUBITAK</td><td>ISO TC164 SC 3 (Hardness testing) and WG 2 (Measurement Uncertainty) meetings are held almost every year. PTB, INRIM and TUBITAK will take part in these meetings and share the project developments and outcomes. The outputs of this project, including the definition of BVK hardness indentations (D1), BVK hardness tests uncertainty models (D4), and required technical specifications for BVK indentation measurement systems (D2) will be disseminated to ISO TC 164 SC3 by PTB, INRIM and TUBITAK. Based on these outputs, a draft work item will be submitted for the revision of ISO 6506-1, 6506-2 and 6506-3 for Brinell, ISO 6507-1, 6507-2 and 6507-3 for Vickers and ISO 4545-1, 4545-2 and 4545-3 for Knoop hardness.</td></tr><tr><td>CIPM CCM WGH</td><td>PTB, INRIM, TUBITAK</td><td>The CIPM CCM Working Group on Hardness (WGH) meets nearly every year to discuss developments in hardness metrology. An important item on the CCM WGH's agenda is the redefinition of hardness scales. A better definition for Brinell, Vickers and Knoop hardness indentations and scientific results related to their measurements will be an important input to this discussion. Project will be shared by the PTB as the chair of the CCM WGH for the current period and by INRIM and TUBITAK as members.</td></tr><tr><td>EURAMET TC-M and WG Hardness</td><td>INRIM, PTB, TUBITAK</td><td>EURAMET TC-M WG Hardness was established to focus on hardness metrology activities in Europe. The progress and outcomes of this project will be placed on the WG meetings' agenda, and the developments in the project will be shared by INRIM, PTB and TUBITAK in every meeting during the course and after the completion of the project.</td></tr><tr><td>COOMET TC-M</td><td>PTB, TUBITAK</td><td>COOMET TC-M meets annually and encompasses the subject of hardness metrology. The progress and outcomes of this project will be shared by PTB and TUBITAK at the COOMET TC-M meetings.</td></tr><tr><td>GULFMET TC-M</td><td>TUBITAK</td><td>GULFMET was established in 2015 and became a fully recognised Regional Metrology Organisation in 2021. The Technical Committee on Mass encompasses hardness metrology and meets twice a year. 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		shared in all TC-M and GA meetings by TUBITAK which are an associate member and regular attendee of GULFMET meetings.	
A5.1.6 M05	ZAG together with TUBITAK will organise and lead one training course of 3 to 4 days for at least one researcher from emerging NMIs (IMBiH and INM) and from other participants (UNZE, BEV-PTP, CMI, ZAG) on current methods of hardness measurements (calibrations). The training will cover the theoretical and practical aspects of the steps involved in the dissemination of the hardness unit.		ZAG, TUBITAK
A5.1.7 M36	A workshop for stakeholders will be organised at the end of the project at TUBITAK. It will be advertised through the project website, newsletters, and the stakeholder committee. The aim of the workshop is to disseminate and demonstrate results from the project. The workshop will present the outputs achieved in the project and will allow time for their discussion with all the participating stakeholders. Targeted attendance is 40+ persons.		TUBITAK, all participants

C5.b Task 5.2: Exploitation and uptake

Activity number	Activity description	Participants (Lead in bold)
A5.2.1 M36	A dissemination, communication and exploitation plan (DCE) will be created at the beginning of the project by TUBITAK with support from all participants and submitted to EURAMET at M6. The DCE plan will be reviewed and updated at least at each project meeting. The exploitation plan will provide further details on the following expected results: - Expected result 1 – Definitions for indentation boundaries in BVK (D1) - Expected result 2 – Precise and comprehensive measurement methodology where required measurement parameters are determined for BVK (D2) - Expected result 3 - BVK indentations transfer standards (D3) - Expected result 4 – Review of hardness testing and calibration machines, implemented national and international standards for industrial applications - Expected result 5 – Uncertainty model for BVK (D4) - Expected result 6 – Guideline document for establishment of traceability chain for BVK (D5) - Expected result 7 – List of recommendations of technical specifications for hardness instrument producers (D6) - Expected result 8 – Software/Algorithm for automation of BVK indentation measurements (D7) - Expected result 9 – Revision of standards for BVK hardness measurement - Expected result 10 – Contribution to EURAMET TC-M/WG on Hardness open access database containing the sensitivity coefficients of the different influence parameters in BVK. - Expected result 11 – Revision of EURAMET cg-16, Version 2.0: Guidelines on the Estimation of Uncertainty in Hardness Measurements. The consortium will ensure the exploitation of results by disseminating the results to standardisation bodies for revision of existing standards, providing defined indentation border artefacts, recommended technical specifications for manufacturers and guidelines for NMIs/Dis, calibration laboratories and other interested end-users.	TUBITAK, all participants
A5.2.2 M36	Artefacts of Brinell, Vickers and Knoop indentations with defined indentation borders developed in the course of the project to serve as transfer standards will be produced by PTB, TUBITAK and INRIM and made available to NMIs, calibration laboratories and to the users of hardness testers.	PTB, all participants
A5.2.3 M36	Recommended technical specifications for improved next generation testing machines for Brinell, Vickers and Knoop hardness measurements (D6) will be distributed to instrument producers via participants' established contacts and the members of the Stakeholder Committee to ensure their widespread uptake by industry.	INRIM, all participants

A5.2.4 M36	Advanced real uncertainty models for Brinell, Vickers and Knoop hardness tests (D4) developed in the project will be shared with NMIs and Dis, accredited calibration laboratories and other interested end-users via established contacts of participants, at meetings of EURAMET TC-M WG Hardness, GULFMET TC-M and CCM WG Hardness.	INRIM , all participants
A5.2.5 M36	Guidelines for traceability in Brinell, Vickers and Knoop hardness measurement developed in the project will be shared with NMIs and Dis, accredited calibration laboratories and other interested end-users via established contacts of participants, at meetings of EURAMET TC-M WG Hardness, GULFMET TC-M and CCM WG Hardness.	INRIM , all participants
A5.2.6 M36	Designs for mechanical modifications to indentation measurement systems and report on software for a fully automatic indentation measuring system for Vickers hardness developed in the project will be shared with NMIs and Dis, accredited calibration laboratories and other interested end-users via established contacts of participants, at meetings of EURAMET TC-M WG Hardness, GULFMET TC-M and CCM WG Hardness.	TUBITAK , all participants
A5.2.7 M36	INRIM, TUBITAK and PTB will submit the report on the validation of advanced uncertainty models developed for BVK hardness tests (D4) and guidelines for the establishment of traceability in Brinell, Vickers and Knoop hardness measurements (D5) developed in the project to EURAMET TC-M for the revision of EURAMET cg-16, version 2.0 (03/2011).	INRIM , TUBITAK, PTB
A5.2.8 M36	INRIM, TUBITAK, CMI and PTB will submit the report summarising the potential effect of various influence factors to EURAMET TC-M/WG-H for developing an open access database containing the sensitivity coefficients to the different influence parameters in Brinell, Vickers and Knoop hardness measurements.	INRIM , TUBITAK, CMI, PTB
A5.2.9 M18, M36	The consortium will identify measures that they will use to demonstrate that the project has narrowed the gap between the capabilities of their consortium and other NMIs/Dis in Europe. Summaries will be produced at months 18 and 36, demonstrating how the project helped less experienced NMIs in the consortium to develop their capabilities. Where appropriate this improvement will be quantified. All participants will provide input to these summaries and the coordinator will provide this information demonstrating the narrowing of the capability gap at the mid-term review and at the end of the project.	TUBITAK , all participants

All IP and potential licencing/exploitation will be handled in accordance with the Grant Agreement and the Consortium Agreement.

C6 WP6: Management and coordination

C6.a Task 6.1: Project management

Activity number	Activity description	Participants (Lead in bold)
A6.1.1 M36	The project will be managed by the coordinator from TUBITAK, who will be supported by the project management board consisting of one representative from each beneficiary, including the leaders of each work package. The members of the project management board will guide the project, attend the project meetings, organise the progress meetings at their local institutes and call additional meetings if needed to ensure the overall project's success.	TUBITAK , all participants
A6.1.2 M36	The work package leaders will report on the on-going progress of the project to the coordinator by e-mail and telephone conferences.	TUBITAK , all participants
A6.1.3 M36	The coordinator, with support from the participants, will manage the project's risks to ensure timely and effective delivery of the scientific and technical objectives and deliverables.	TUBITAK , all participants
A6.1.4 M36	The consortium will ensure that any ethics issues identified are addressed.	TUBITAK , all participants

C6.b Task 6.2: Project meetings

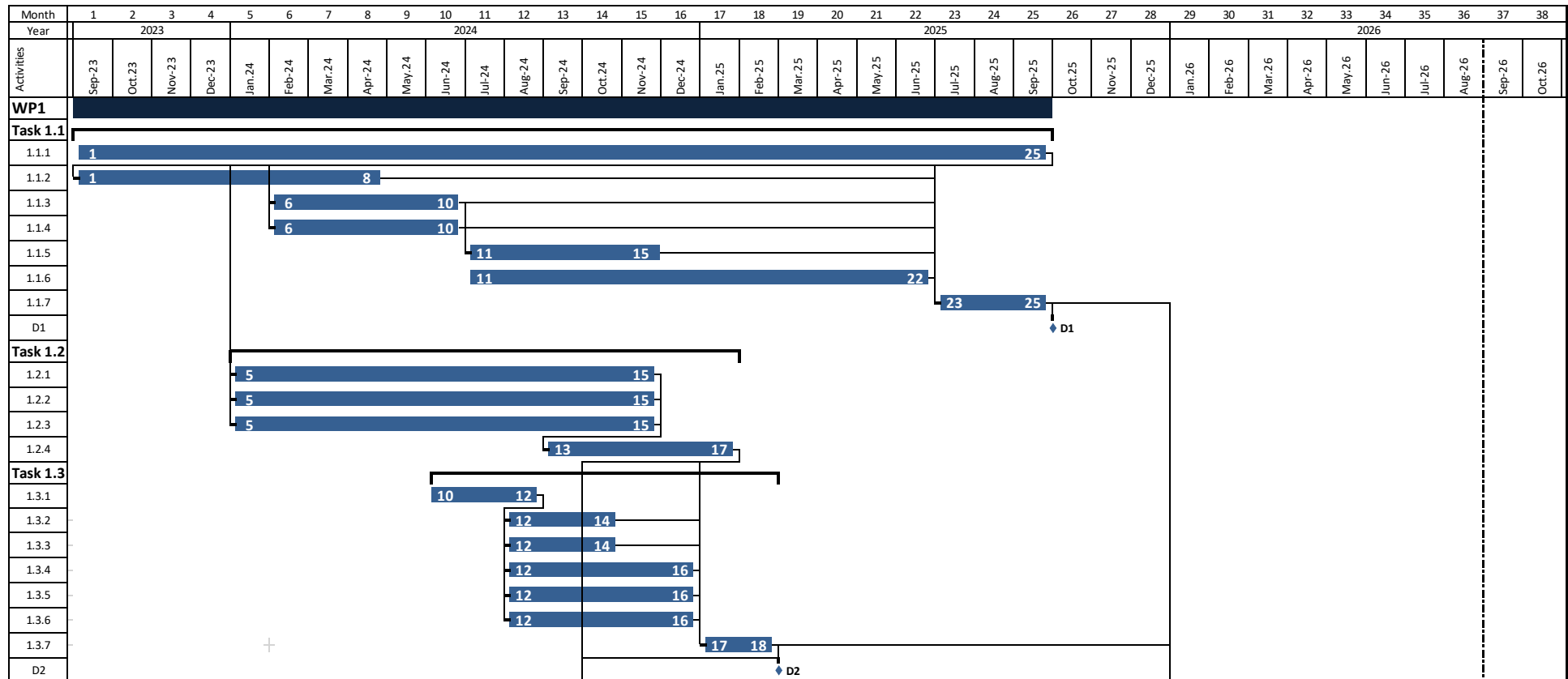
Activity number	Activity description	Participants (Lead in bold)
A6.2.1 M2	The kick-off meeting involving all participants will be held approximately within two months after the start of the project at TUBITAK.	TUBITAK , all participants
A6.2.2 M36	There will be five formal project meetings. These meetings include the kick-off, mid-term (around M18) and final meeting (around M36). In addition, two further meetings will be held around M9 and M27. The meetings will be held prior to reporting. The meetings will review progress and will be used to ensure participants are clear as possible concerning their responsibilities in the subsequent period. The location of the meetings will rotate among the participants. Where possible, meetings may be held as satellite meetings to important conferences or committee meetings.	TUBITAK , all participants
A6.2.3 M36	In addition to the formal project meetings, technical meetings of work package groups may be held whenever necessary and will be arranged on an ad-hoc basis.	TUBITAK , all participants

C6.c Task 6.3: Project reporting

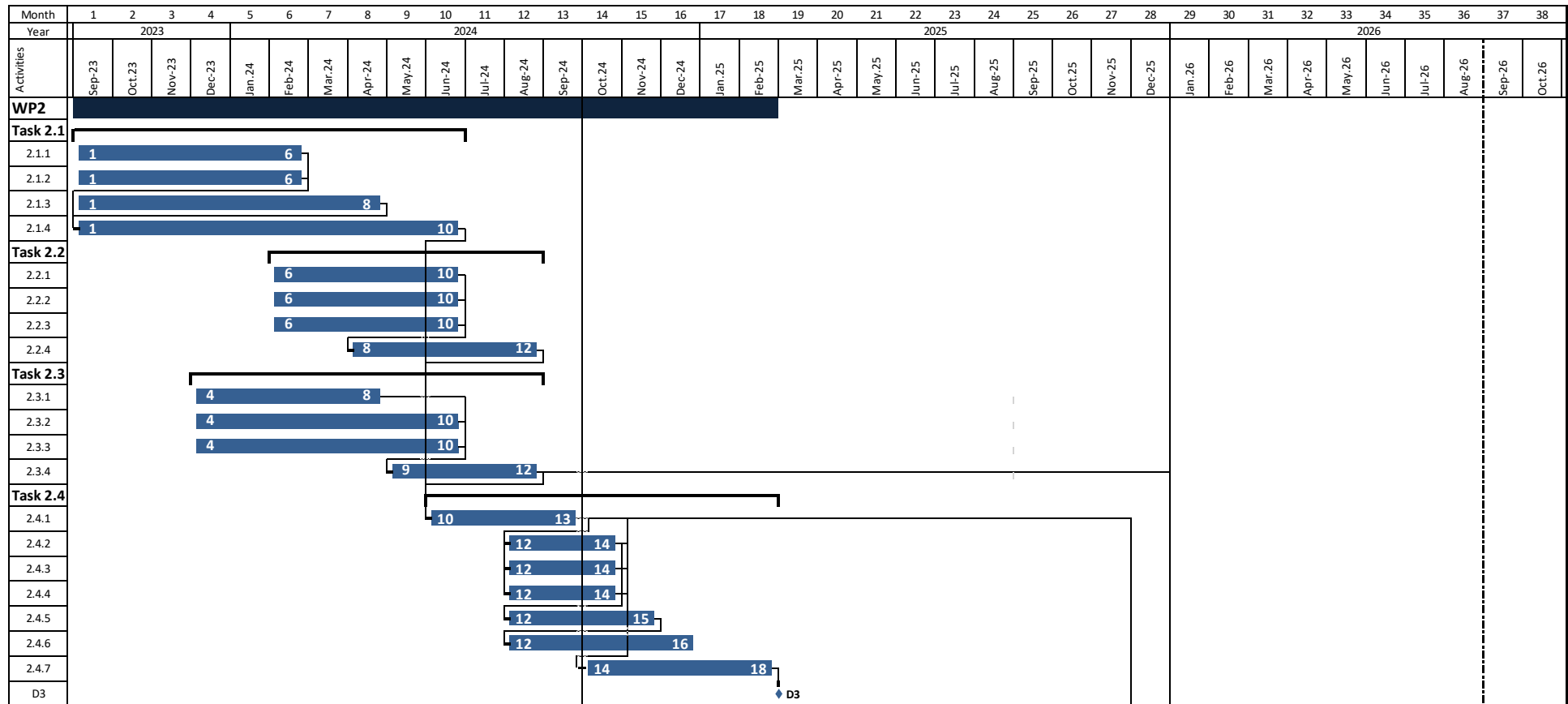
Activity number	Activity description	Participants (Lead in bold)
A6.3.1 M1	One month after the start of the project a publishable summary will be produced and submitted to EURAMET.	TUBITAK , all participants
A6.3.2 M6	Six months after the start of the project a data management plan (DMP) and a dissemination, communication and exploitation plan (DCE) will be produced and submitted to EURAMET.	TUBITAK , all participants
A6.3.3 M36 +60 days	Following Articles 19 and 21 and the data sheet of the grant agreement, information will be submitted to EURAMET, in accordance with the procedures issued by them to enable EURAMET to comply with its obligations to report on the programme to the European Commission. • Progress reports will be submitted at months 9 and 27 (May 2024, November 2025 + 45 days); 18 and 36 (February 2025, August 2026 + 60 days). • Outcomes and Impact reports and updated publishable summaries will be submitted at the same times. All participants will provide input to these reports and the coordinator will provide these and updated publishable summaries to EURAMET. Where necessary, additional reports and / or information may be requested to enable EURAMET to comply with its obligations to the European Commission.	TUBITAK , all participants
A6.3.4 M36 +60 days	Periodic Reports (including financial reports, updated data management plan, and updated dissemination, communication and exploitation plan) will be delivered at months 18 (February 2025 + 60 days) and 36 (August 2026 + 60 days) in accordance with Articles 19 and 21 and the data sheet of the grant agreement. All participants will provide input to these reports and the coordinator will provide these to EURAMET.	TUBITAK , all participants
A6.3.5 M36 +60 days	Final report will be delivered at month 36 (August 2026 + 60 days) in accordance with Articles 19 and 21 and the data sheet of the grant agreement. All participants will provide input to these reports and the coordinator will provide these to EURAMET.	TUBITAK , all participants
A6.3.6 M19	The project will be subject to a midterm review in Spring 2025. Reports (project self-assessment, updated publishable summary and presentation) will be delivered prior to the midterm reviews for Call 2022, following the schedule detailed by EURAMET for the specific review. All participants will provide input to these reporting documents and the coordinator will provide the documents to EURAMET.	TUBITAK , all participants

Formal reporting will be in line with EURAMET's requirements and will be submitted in accordance with the Reporting Guidelines.

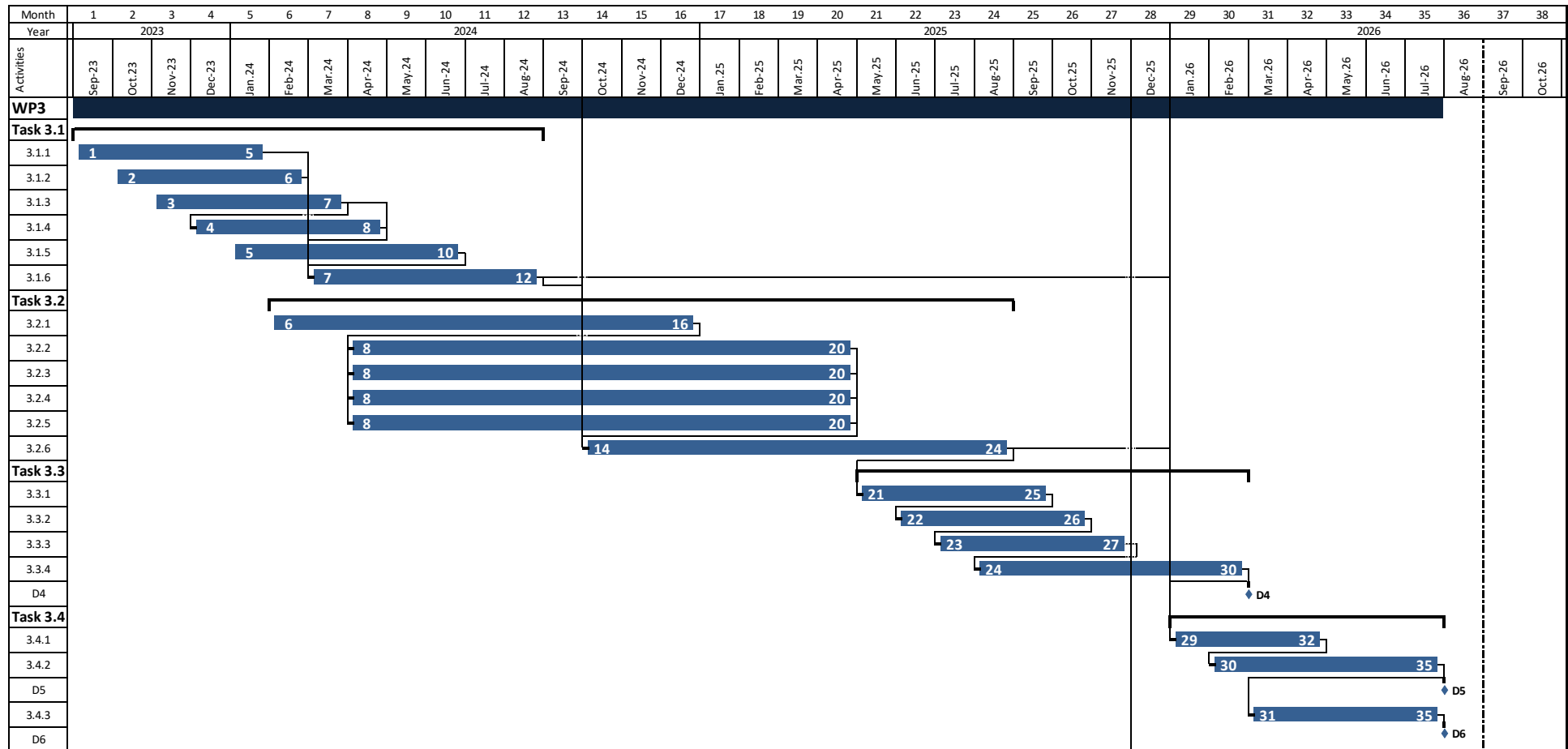
C7 Gantt chart



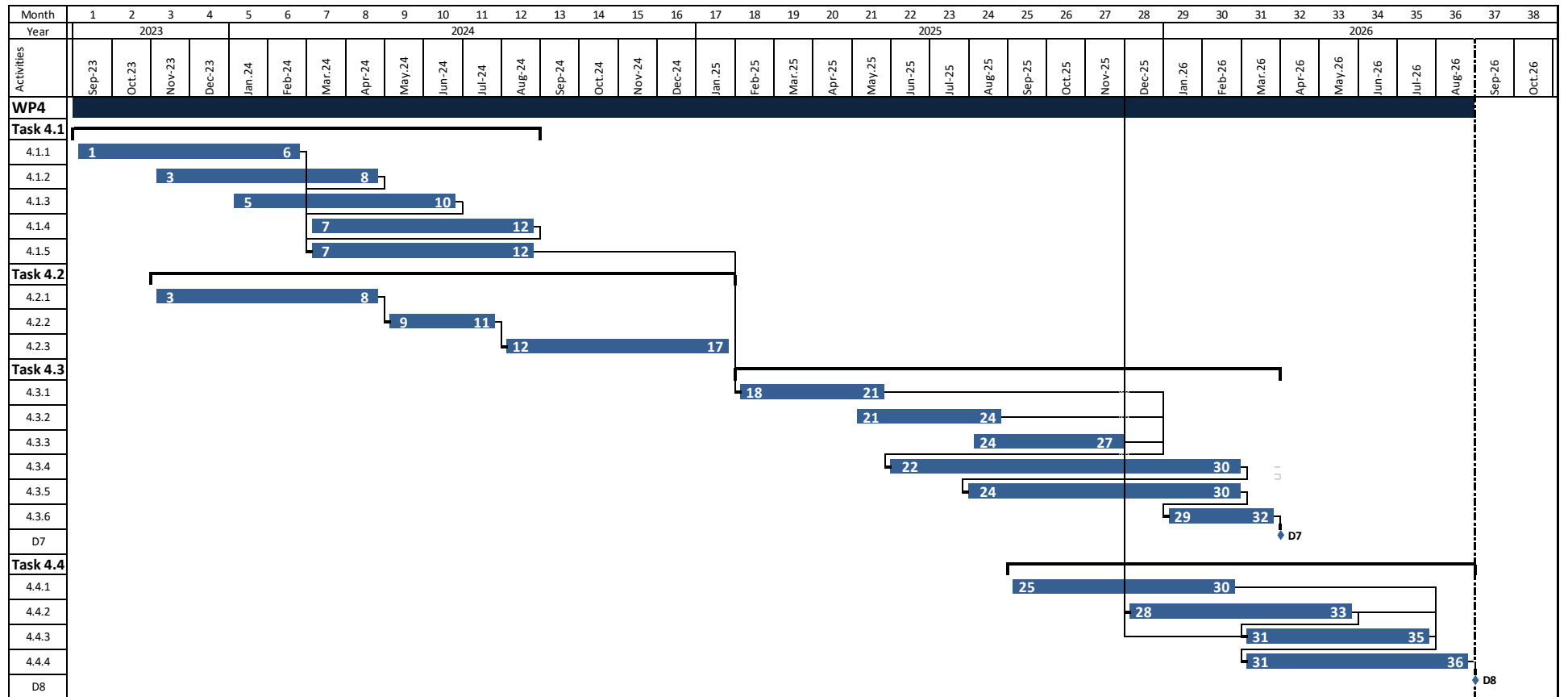
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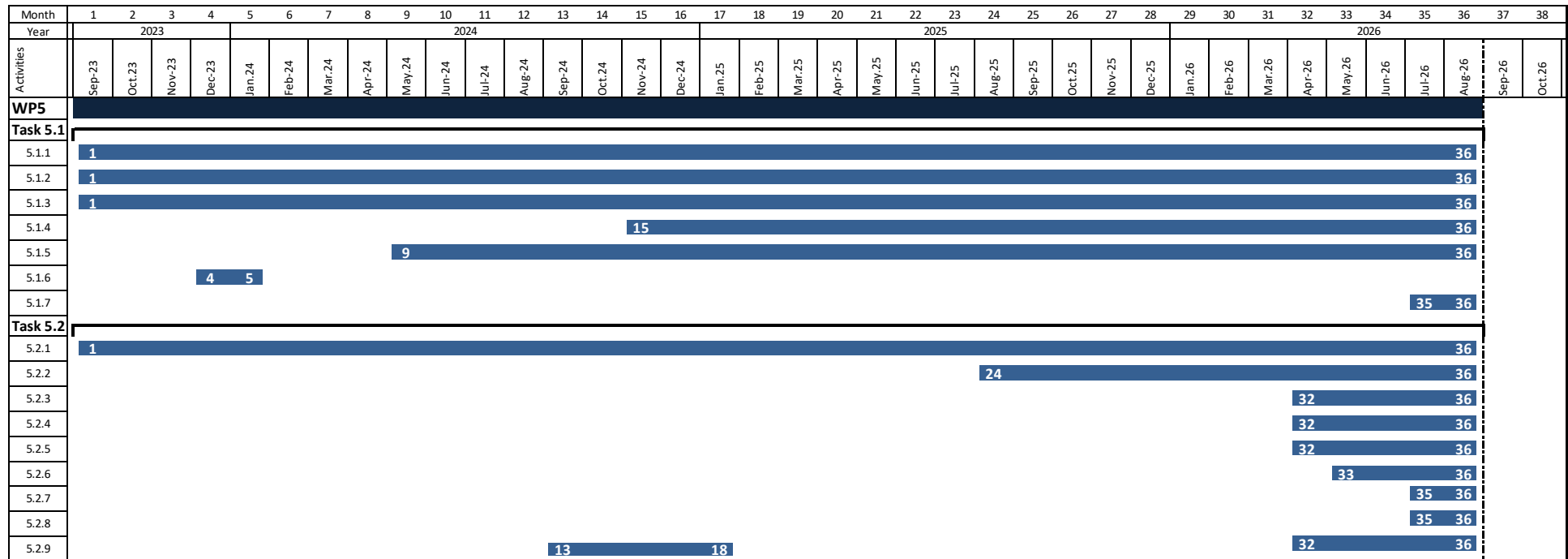
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European Partnership on Metrology



European Partnership on Metrology



European Partnership on Metrology

Month	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38
Year	2023				2024												2025												2026									
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Section D: Risk and risk mitigation

D1 Scientific/technical risks

Risk (description)	Likelihood, impact and severity of occurrence	Mitigation i.e. what the consortium will do to decrease the likelihood of the risk occurring	Contingency i.e. what the consortium will do if despite the mitigation the risk still occurs
Task 1.1: Lack of machine availability	Likelihood after mitigation: Low Impact: The measurement schedule would be delayed. Level of severity: Low	The participant will ensure that the measurement plan is flexible and can be modified according to machine availability.	This task can be rescheduled so that the measurements can be made later in the project period without having a significant knock-on effect on other activities.
Task 1.2: Variation in parameters such as illuminance, focal point etc. cannot be set fine enough	Likelihood after mitigation: Low Impact: The measurement schedule would be delayed. Level of severity: Low	By using different devices and portioning between different participants, the best possible devices can be used here.	Measurement tasks must be distributed to other participants.
Task 1.3: Not all lighting options can be implemented	Likelihood after mitigation: Low Impact: The measurement schedule would be delayed. Level of severity: Low	Due to the large number of participants, different influencing factors can be investigated.	Measurement tasks will be distributed to other participants.
Task 2.1: Material stability for soft materials may not be good enough to manage reference indentations	Likelihood after mitigation: Medium Impact: The indentation size may change and may create a lack of consistency problem between the participants. Level of severity: Medium	Different materials will be investigated in order to assure selection of those with sufficient stability.	Calibration period of soft material with insufficient stability will be shortened and possible deviations in the indentation size will be accounted for.
Task 2.4: Quality of indentations realised may not be sufficient to serve as reference indentations	Likelihood after mitigation: Medium Impact: Higher uncertainty in the measurement results. Level of severity: Medium	Different machines will be used for realisation of indentations.	The indentations will be realised with different identical materials at different NMIs, and the best indentation will be selected as the reference indents.
Task 3.1: Due to confidentiality issues, there could be a problem obtaining information about the stakeholders claimed capabilities	Likelihood after mitigation: Very low Impact: Inability to understand stakeholder needs. Level of severity: Low	It should be possible to give strictly anonymised information about the data in this task.	Data for the necessary database can be collected from the knowledge of the participants to nearly the same level.
Task 3.2: Lack of machine availability	Likelihood after mitigation: Very low Impact: The measurement schedule would be delayed. Level of severity: Low	The consortium will ensure that the measurement plan is flexible and can be changed regarding machine availability.	Activities will be redistributed to participants with access to needed instruments.
Task 3.3: The advanced uncertainty model for Brinell, Vickers and Knoop hardness tests is unable to be validated due to a lack of data	Likelihood after mitigation: Low Impact: The project will not be able to validate all the uncertainty contributions as expected. Level of severity: Medium	Additional measurements will be performed to generate additional data which can then be validated. If necessary, the measurement uncertainty will be increased to obtain consistency between the model and the data.	If the discrepancy between the measurement results is larger than the corresponding uncertainty, then additional data from the participants will be used.

Task 4.2: Inability to find a marker system to locate the indentation region on surface of the block	Likelihood after mitigation: Medium Impact: Automatisations of indentation location marking will not be possible. Level of severity: Low	The marker apparatus will be designed, produced and mounted to the system.	Marking of the indentation locations will be made semi-automatic with the help of the operator.
Task 4.3: Inability to perform automatic indentation measurements for many types of indents according to the new definition that will be created in WP1	Likelihood after mitigation: Medium Impact: A significant number of non-measurable indents. Level of severity: Medium	Semi-automatic measurements will be performed where the border of indentation will be decided by the operator.	A manual measurement method will be developed with better repeatability with different operators.

D2 Management risks

Risk (description)	Likelihood, impact and severity of occurrence	Mitigation i.e. what the consortium will do to decrease the likelihood of the risk occurring	Contingency i.e. what the consortium will do if despite the mitigation the risk still occurs
Key personnel are lost to the project	Likelihood after mitigation: Low Impact: The loss of key team members would create difficulties in delivering the project, or specific tasks or deliverables. Level of severity: Low	None of the team members are planning to leave or retire within the project. The grouping of experts within the consortium should minimise the areas where knowledge is held by a single person. All the participants will identify backups for key workers wherever possible to reduce the overall risk to the project. Project plans will be shared within the consortium and results and methodology will be documented.	If a key member leaves the project, then the participant concerned will be responsible for appointing a replacement first preference from the same participant and field of work. However, this may still lead to a delay in delivery.
Complexity of managing a large consortium	Likelihood after mitigation: Medium Impact: Failure to fully cooperate or communicate effectively within the consortium could endanger efficient delivery of the project. Level of severity: Low	The participants are all experienced with complex multinational projects. Many have previously developed close relationships through collaborating within other European consortia. Regular communication and feedback will ensure that potential problems are identified early and that all participants are clear on their roles.	WP leaders will play an important role in flagging up potential problems to the coordinator and the project management board, who will then decide on the best course of action to take. If necessary, work will be reassigned to an alternative participant, or parts of the work re-scoped in agreement with EURAMET.
Inter-dependencies between technical activities and tasks are too complex	Likelihood after mitigation: Medium Impact: Tasks are delayed, or it is not possible to deliver them. Level of severity: Low	Technical meetings run by the WP leaders have been scheduled to ensure proper sharing of knowledge. The interdependencies between tasks will be considered at meetings to ensure that this is addressed properly in the planning of the work. The technical WPs will be closely managed by their WP leaders to ensure that they deliver their own outputs.	In most cases, activities on the critical path have some overlap in time and thus a delay in the output of one deliverable does not necessarily cause an immediate delay in another.

Problems dealing with Intellectual Property (IP) ownership and/or exploitation might occur and could be a source of potential conflict	Likelihood after mitigation: Low Impact: Disagreement between the participants could delay the progress of the project, i.e. in realisation of activities. Level of severity: Low	All beneficiaries will sign the grant agreement and all participants the consortium agreement, which includes IP clauses. IP will be handled accordingly.	Independent arbitrators will be used in the event of disagreement between participants.
The onsite facilities of participants, and/or access to public/commercial services or sites is restricted for a period of time during the project due to an extraordinary event or situation that is beyond the participants' control e.g. COVID-19	Likelihood after mitigation: High Impact: Activities and deliverables are delayed, or no longer able to be completed. Level of severity: High	In most cases, activities on the critical path have been scheduled to have some overlap in time and thus a delay in the output of one activity will not necessarily cause an immediate delay in another.	Where possible, work will be reassigned to an alternative participant, or re-planned, therefore minimising delays and technical deviations that would have a negative impact on the project. Adaptive measures will be activated.

D3 Ethics

The Partnership Ethics Review 2022 has given JRP 22RPT01 TraInd BVK-H "Ethics clearance".

Data protection

By signing or acceding to this grant agreement and / or consortium agreement each participant asserts that the requirements of the General Data Protection Regulation (GDPR) 2016/679 which entered into force on 25 May 2018 will be met. Under the regulation, the data controllers and processors are fully accountable for the data processing operations. Any violation of the data subject rights may lead to sanctions as described in Chapter VIII, art.77-84 of the GDPR.

If personal data are transferred from the EU to a non-EU country or international organisation, such transfers will be in accordance with Chapter V of the GDPR 2016/679. If personal data are transferred from a non-EU country to the EU (or another third state), such transfers will comply with the laws of the country in which the data was collected.

Non-EU countries

The consortium will ensure that participants and collaborators, including those from non-EU countries, fully adhere to Horizon Europe ethics standards and guidelines, no matter where the research or activities are carried out and that research or activities performed outside the European Union are compatible with EU, national and international legislation and can be legally conducted in one of the EU Member States. If applicable, details on the material, samples and/or equipment which will be imported to/exported from EU must be provided and the adequate authorisations granted by the relevant authorities have been or will be obtained and kept on file by the consortium. The consortium will also, in the case of dual use applications, clarify whether any export licence is required for the transfer of knowledge, equipment or material.

Ethical integrity

The participants will ensure that all ethics issues related to activities in the project are addressed in compliance with ethical principles (including the highest standards of research integrity as set out in the ALLEA European Code of Conduct for Research Integrity https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/european-code-of-conduct-for-research-integrity_horizon_en.pdf), the applicable international and national law, and the provisions set out in the grant agreement. This includes the ethics issues identified in the ethics screening and the submitted documents, and any additional ethics issues that may emerge in the course of the project. In the case where any substantial new ethics issues arise, participants will inform the granting authority EURAMET e.V, and for each ethics issue applicable, participants will follow the guidance provided in the Horizon Europe 'How to complete your ethics self-assessment' guide'.

The consortium will ensure that appropriate procedures, policies and structures (https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/guideline-for-promoting-research-integrity-in-research-performing-organisations_horizon_en.pdf) are in place to foster responsible research practices, to prevent questionable research practices and research misconduct, and to handle allegations of breaches of the principles and standards in the Code of Conduct.

Section E: References

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